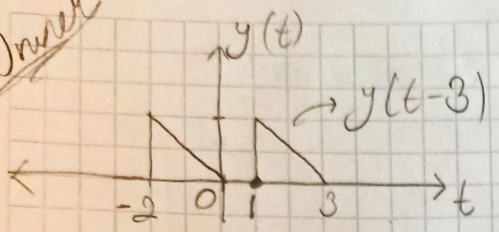
Örnek



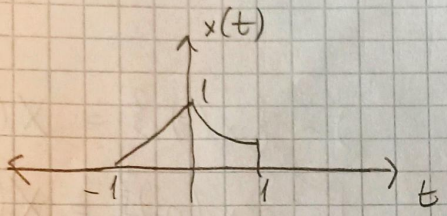
Genel formül

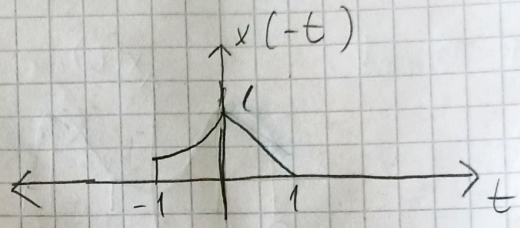
$x(t-a)$

$a<0$ ise sola

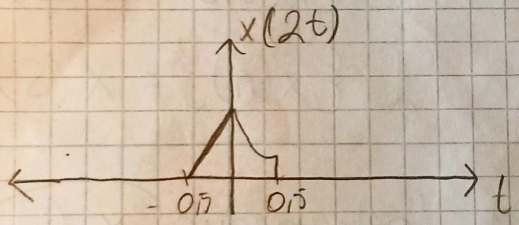
$a>0$ ise sağa

Zamanda ters çevirme



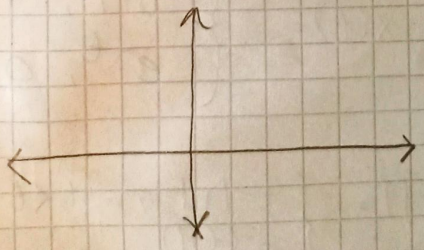


Zamanda ölçekleme



t	x(t)	x(2t)
-1	0	0
-0.5	0.5	0
0	1	1
0.5	$1/\sqrt{e}$	$1/e$
1	$1/e$	0

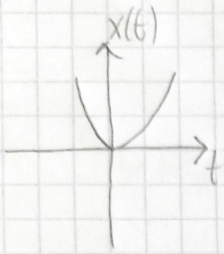
Kaydırma ve ölçekleme



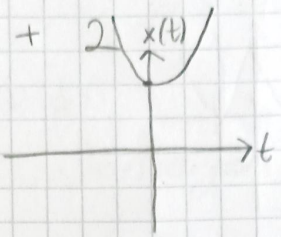
$x(at-b)$

1. Önce zamanda kaydır.
   $x(t-b)$
2. Sonra ölçekle
   $x(at-b)$

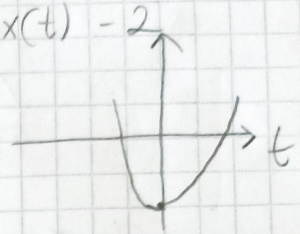
Genlik Ölçekleme ve Kaydırma



$x(t) + 2$



$x(t) - 2$



Simetri

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{1}{2}\left(x(t) + x(-t)\right)$$

$$x_o(t) = \frac{1}{2}\left(x(t) - x(-t)\right)$$

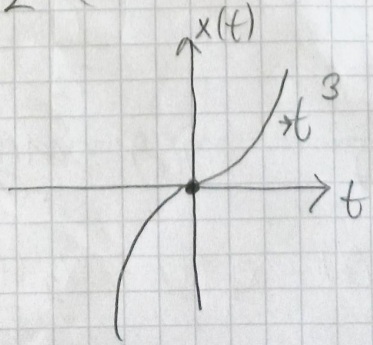
Örnek: $x(t) = t^2 + 1$

$$x_e(t) = \frac{1}{2}\left(t^2+1 + (-t)^2+1\right) = t^2+1$$

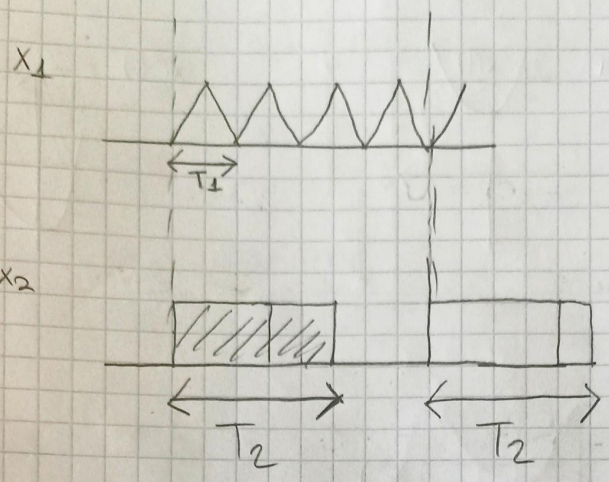
$$x_o(t) = \frac{1}{2}\left(t^2+1 - \left((-t)^2+1\right)\right) = 0$$

Örnek: $x(t) = t^3$

$$x_o(t) = \frac{1}{2}\left(t^3 - (-t)^3\right) = t^3$$



Periyodiklik



$\frac{T_1}{T_2} \in \mathbb{Q}$ ise $x_1 + x_2$ periyodiktir.

Örnek: $x_1(t) = \sin \pi t$

$x_2(t) = \sin t$

$y(t) = x_1(t) + x_2(t)$

$x_1 \rightarrow T_1 = \frac{2\pi}{\pi} = 2$

$x_2 \rightarrow T_2 = \frac{2\pi}{1} = 2\pi$

$\frac{2}{2\pi} = \frac{1}{\pi} \notin \mathbb{Q}$ periyodik değildir.

Örnek: $x_1(t) = \cos(2\pi t + \frac{\pi}{4})$

$x_2(t) = \sin 7\pi t$

$x_1 \rightarrow T_1 = \frac{2\pi}{2\pi} = 1$

$x_2 \rightarrow T_2 = \frac{2\pi}{7\pi} = \frac{2}{7}$

$\frac{T_1}{T_2} = \frac{7}{2} \in \mathbb{Q}$

$y(t)$ periyodiktir. periyodu nedir?

$y(t) = x_1(t) + x_2(t)$

$2T_1 = 7 \cdot T_2 = T = 2s$

x fonksiyonu için

a) $q(t) = x(t-3)$

q nedenseldir.

b) x tek fonksiyon yani $x(t) = -x(-t)$

hangi t değerleri için $x = 0$ 'dır?

$q(t) = 0 \quad , \quad t < 0$

$q(t) = x(t-3) \quad , \quad t < 0$

$t = t+3$ için

$x(t) = 0 \quad t < -3$

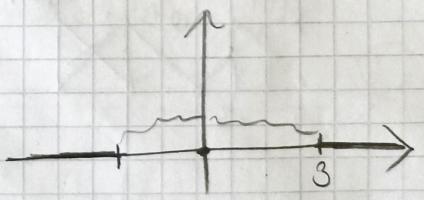
$x(t) = -x(-t)$

$-x(-t) = 0 \quad , \quad t < -3 \qquad t = -t$ için

$-x(t) = 0 \quad , \quad -t < -3 \qquad x(t) = 0 \quad , \quad t = 0$

$-x(t) = 0 \quad , \quad t > 3$

$$x(t) = \begin{cases} x(t) = 0 & t < -3 \\ x(t) = 0 & t > 3 \\ x(t) = 0 & t = 0 \end{cases}$$



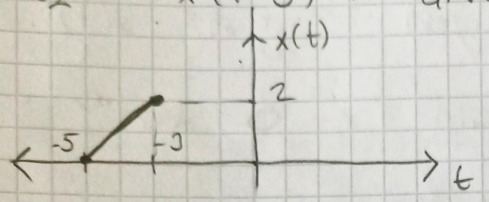
Soru: x için;

a) $x(t) = t + 5$, $-5 \leq t \leq -3$

b) $q_1(t) = x(t-5)$: nedensel

c) $q_2(t) = x(t-3)$: çift fonksiyon

a)



x(t)

b) $q_1(t) = x(t-5) = 0$, $t < 0$

$x(t) = 0 \Rightarrow t+5 < 0$

$t < -5$

$x(t) = x(-t)$

$t = -t$ için yani

c) $q_2(t) = q_2(-t)$

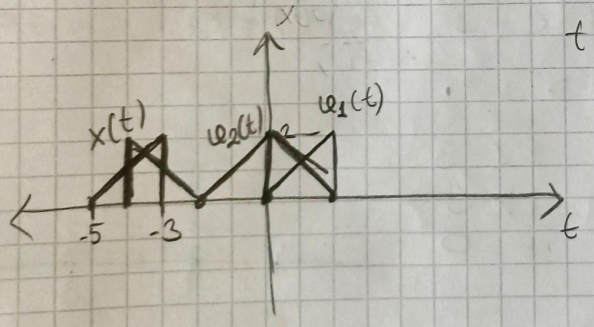
$x(t-3) = x(-t+3)$

$-(t-3)$

$-t+3$

$t+5$

$q_2(t) = x(-t-3)$



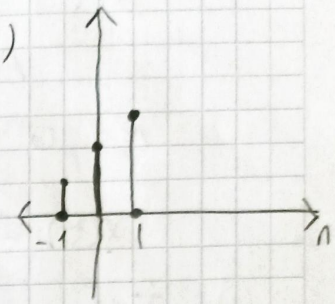
x bir dizi olmak üzere

a) $x(n) = n+2$ $\quad -1 \leq n \leq 1$

b) $u_1(n) = x(n-1)$; nedensel

c) $u_2(n) = x(n+1)$; çift fonksiyon

a)



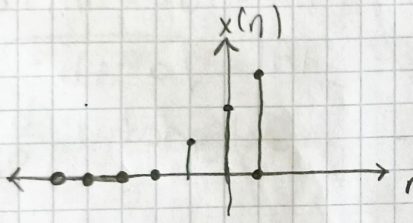
$u_1(n) = x(n-1) = 0 \quad , n < 0$

$n = n+1$ için

b) $u(n+1) = x(n) = 0 \quad n < -1$

$n+1 < 0$

$x(n)=0, n < -1$



c) $u_2(n) = u_2(-n)$

$\boxed{x(n+1) = x(-n+1)}$

$n = n-1$ için

$x(n) = x(-n+1)$

$x(n) = x(-n+2)$

$x(n) = n+2 \quad , -1 \leq n \leq 1$ (1)

$x(-n+2) = -n+4 \quad , -1 \leq -n+2 \leq 1$

$= -n+4 \quad , 1 \leq n \leq 3$

$x(-n+2) = -n+2+2$

$= -n+4$

Karmaşık Sinüzoidler

$$x(t) = A \cdot e^{\lambda t}$$

$$A, \lambda \in C$$

$$\lambda = \sigma + j\omega \qquad \sigma = 0 \Rightarrow \lambda = j\omega$$

$$x(t) = A \cdot e^{j\omega t}$$

$$e^{jt} = \cos t + j \sin t$$

$$e^{-jt} = \cos t - j \sin t$$

$A \in C \Rightarrow$ yani $A = a + jb$ ise

$$A = a + jb = |A| \cdot e^{j\theta}$$

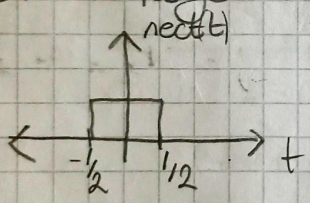
$$|A| = \sqrt{a^2 + b^2}$$

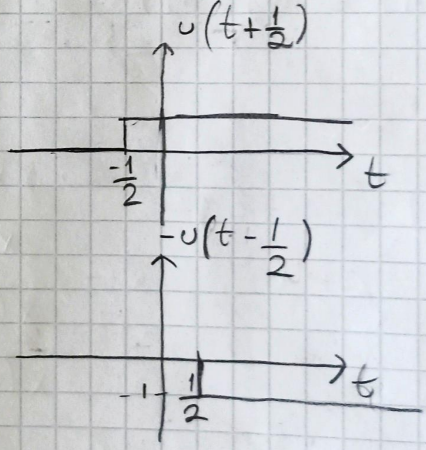
$$\theta = \arctan\left(\frac{b}{a}\right)$$

$$A \in C \Rightarrow |A| \cdot e^{j\theta} \cdot e^{j\omega t}$$

$$= |A| \cdot e^{j(\theta + \omega t)}$$

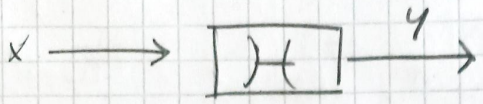
Dikdörtgen Fonksiyonu





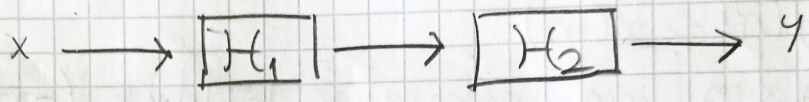
Sistemlerin Özellikleri

Sürekli ve Ayrık Zamanlı Sistemler



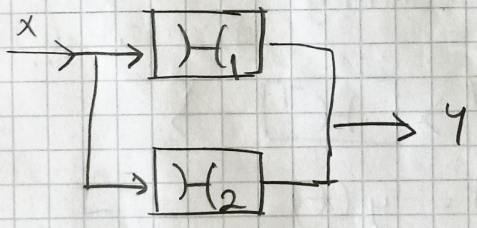
$$y = \mathcal{H}x$$

Seri sistem bağlantısı



$$y = H_2 H_1 x$$

Paralel sistem bağlantısı



$$y = \mathcal{H}_1 x + \mathcal{H}_2 x$$

Sistem Belleği

$y(t_0) = \mathcal{H}x(t_0 - 5)$ } Bellekli sistem

$y(t) = \mathcal{H}x(t + t_0)$ }

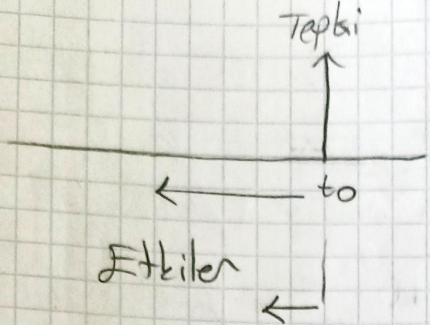
$y(t_0) = \mathcal{H}x(t_0)$ } Bellekiz sistem

$y(n) = \mathcal{H}x(n) = x(-n)$ } Bellekli sistem

$y(t) = \mathcal{H}x(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$ } Bellekli sistem

$\mathcal{H}x(n) = e^{x(n)}$ } Bellekiz sistem

Sistem Nedenselliği



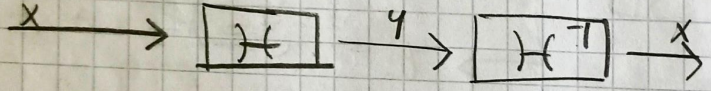
Örnekler: $y(n) = x(-n)$

$n < 0$ için nedensel değildir (anti-nedensel)

$n \geq 0$ için nedenseldir

$H(x(n)) = (n+1).\, e^{x(n-1)}$ nedenseldir. Her $x \in \mathbb{R}$

Sistemlerin Tersine Çevrilebilirliği



Örnek: $H(x(t)) = x(t-t_0) = y(t)$

$t = t + t_0$ için

$H(x(t+t_0)) = x(t) = \underline{y(t+t_0)}$

$H^{-1} y(t)$

Örnek: $H(x(n)) = 3\, x(3n+3)$

$n = \frac{1}{3} n - 1$ için

$$H\left(x\left(\frac{1}{3}n - 1\right)\right) = \frac{1}{3}\, y\left(\frac{1}{3}n - 1\right) = H^{-1} y(n)$$

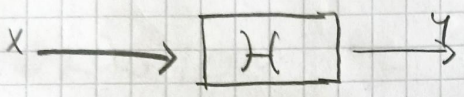
Sistemlerin BIBO kararlılığı

$y(t) = 2x(t)$ Kararlıdır.

$y(n) = e^{x(n)}$ Kararlıdır.

δ önemli olan genlik kısmıdır.

Sistemlerin Zamanla Değişmezlik



$x(t) \longrightarrow y(t)$

$x(t-t_0) \longrightarrow y(t-t_0)$

$x(t+t_0) \longrightarrow y(t+t_0)$

Örnek: $Hx(t) = \sin x(t)$

1. adım

$x'(t) = x(t-t_0)$

$Hx'(t) = \sin x'(t)$

$Hx'(t) = \sin x(t-t_0)$

$Hx(t-t_0) = \sin x(t-t_0)$

2. adım

$t = t-t_0$ için

$Hx(t-t_0) = \sin x(t-t_0)$

Zamanla değişmez sist.

Örnek: $Hx(t) = x(2t)$

1. adım

$x'(t) = x(t-t_0)$

$Hx'(t) = x'(2t)$

$x'(2t) = x(2t-t_0)$

$Hx(t-t_0) = x(2t-t_0)$ ①

2. adım

$Hx(t-t_0) = x[2\cdot(t-t_0)]$

$= x(2t-2t_0)$ ②

① ≠ ② Zamanla değişmez sist değil

Sistemlerde Doğrusallık

1- Toplamsallık

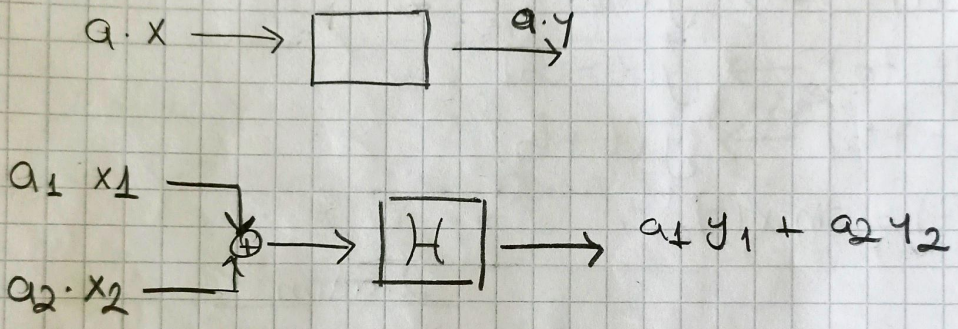
$y = f(x)$

$k \cdot y = f(k \cdot x)$

$y_1 = f(x_1)$ $\qquad y_1 + y_2 = f(x_1) + f(x_2)$

$y_2 = f(x_2)$

2- Homojenlik



Örnek: $H(x(t)) = t \cdot x(t)$

(1) $x'(t) = a_1 \cdot x_1 + a_2 x_2$

$H(x'(t)) = t \cdot x'(t) = t\,[a_1 x_1 + a_2 x_2]$ (1)

(2) $a_1 H(x_1(t)) = a_1\, t\, x_1(t)$

$+\ a_2 H(x_2(t)) = a_2\, t\, x_2(t)$

$a_1 H(x_1(t)) + a_2 H(x_2(t)) = a_1\, t\, x_1(t) + a_2\, t\, x_2(t)$ (2)

1 = 2

Doğrusaldır.

Örnek: $Hx(n) = 2x(n) + 3$ $\quad 1 \neq 2$

Doğrusal değildir.

① $x'(n) = a_1 x_1 + a_2 x_2$

$Hx'(n) = 2.(a_1 x_1 + a_2 x_2) + 3$ ①

② $a_1 H(x_1(n)) = a_1((2.x_1(n)) + 3) = 2a_1 x_1(n) + 3a_1$

$+a_2 H(x_2(n)) = a_2(2x_2(n) + 3) = 2a_2 x_2(n) + 3a_2$

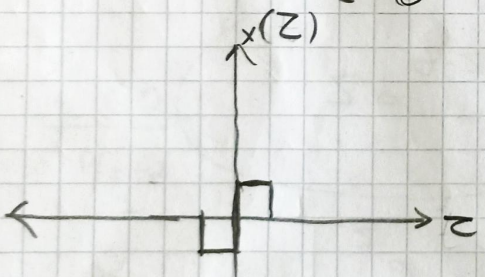
② $= 2a_1 x_1(n) - 3a_1 + 2a_2 x_2 + 3a_2$

KONVOLÜSYON

$$x * h = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$$

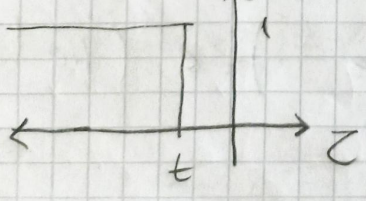
Örnek:

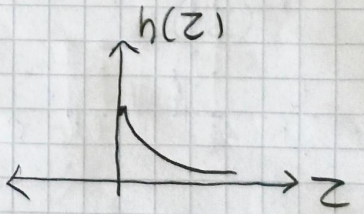
$$x(\tau) = \begin{cases} -1, & -1 \leq \tau < 0 \\ 1, & 0 \leq \tau < 1 \\ 0, & \text{diğer} \end{cases}$$

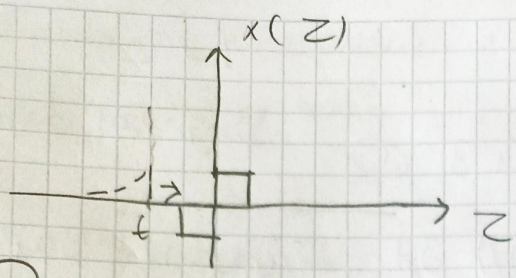


$h(t) = e^{-t} \cdot u(t)$

$x(t) * h(t) = ?$







① $t < -1$ için

$$x * h = \int_{-\infty}^{t} x(z)h(t-z)dz = 0$$

② $-1 < t < 0$ için

$$\int_{-1}^{t} -1 \cdot e^{z-t} dz$$

$$-\left(\int_{-1}^{t} e^{z} dz + \int_{-1}^{t} e^{-t} dz \right)$$

$$= -e^{-t} e^{z} \Big|_{-1}^{t} = -1 + e^{-t-1}$$

③ $0 < t < 1$ için

$$x * h = -\int_{-1}^{0} e^{z-t} dz + \int_{0}^{t} e^{z-t} dz$$

$$= -e^{-t} \cdot e^{z} \Big|_{-1}^{0} + e^{-t} \cdot e^{z} \Big|_{0}^{t}$$

$$-e^{-t} + e^{-t-1} + 1 - e^{-t}$$

④ $t > 1$ için

$$x * h = -\int_{-1}^{0} e^{z-t} dz + \int_{0}^{1} e^{z-t} dz$$

$$-e^{-t} \cdot e^{z} \Big|_{0}^{1}$$

$$-e^{-t} + e^{-t-1} + 1 - e^{-t} + \left(-e^{-t+1} - e^{-t}\right) = e^{-t}\left(-1 + e^{-1} + e - 1\right)$$

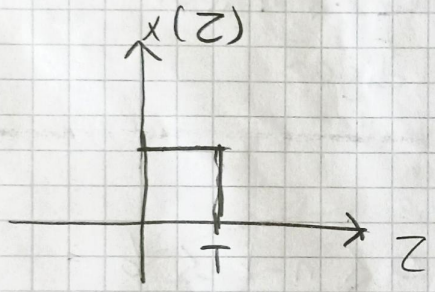
$$x * h = \begin{cases} 0, & t < -1 \\ e^{-t-1} - 1, & -1 < t < 0 \\ (e^{-1} - 2)e^{-t} + 1, & 0 < t < 1 \\ (-2 + e + e^{-1}) \cdot e^{-t}, & t > 1 \end{cases}$$

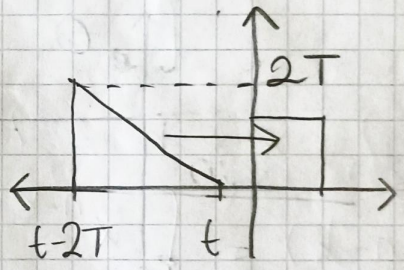
Örnek: $x(t) = \begin{cases} 1 & , 0 \le t \le T \\ 0 & , \text{diğer} \end{cases}$

$h(t) = \begin{cases} t & , 0 \le t \le 2T \\ 0 & , \text{diğer} \end{cases}$

$y = x * h(t) = ?$

$h(t-\tau) = \begin{cases} t-\tau & , 0 \le t-\tau \le 2T \\ 0 & , \text{diğer} \end{cases}$





① $t \le 0$

$y = 0$

② $0 \le t \le T$

$y = \int_0^t (t-\tau)\,d\tau$

$= \left(t\tau - \frac{\tau^2}{2}\right)\Big|_0^t$

$= \frac{t^2}{2}$

③ $T \le t \le 2T$

$y = \int_0^T (t-\tau)\,d\tau = t\tau - \frac{\tau^2}{2}\Big|_0^T$

$= Tt - \frac{T^2}{2}$

④ $2T \le t \le 3T$

$y = \int_{t-2T}^{T} (t-\tau)\,d\tau = t\tau - \frac{\tau^2}{2}\Big|_{t-2T}^{T}$

$= -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2$

⑤ $t > 3T$

$y = 0$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad |a| < 1$$ } Sonsuz Toplam formülü

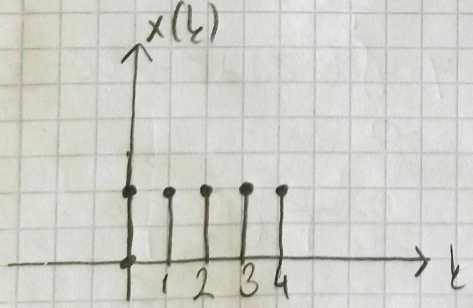
$$\sum_{n=0}^{N} a^n = \frac{1-a^{N+1}}{1-a}$$ } Sonlu toplam formülü

Örnek: $x(n) = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{diğer} \end{cases}$ $\quad a > 1$

$h(n) = \begin{cases} a^n, & 0 \le n \le 6 \\ 0, & \text{diğer} \end{cases}$

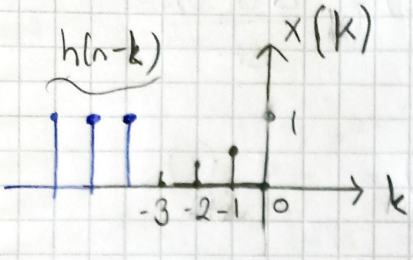
$y(n) = x * h = ?$

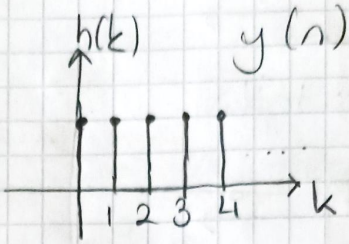
$h(n-k) = \begin{cases} a^{n-k}, & 0 \le n-k \le 6 \\ 0, & \text{diğer} \end{cases}$



Örnek: $x(n) = 2^n \cdot u(-n)$ $\quad h(n) = u(n)$

$y(n) = x(n) * h(n) = ?$





① $k = -k$ için $\quad k = \ell - n$ için

$$y(n) = \sum_{k=-\infty}^{n} 2^k = \sum_{k=\infty}^{-n} 2^{-k} = \sum_{k=-n}^{\infty} 2^{-k} \Rightarrow \sum_{\ell=0}^{\infty} 2^{-(\ell-n)}$$

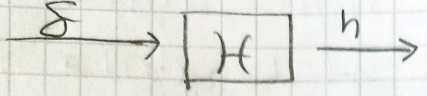
$$= 2^n \sum_{\ell=0}^{\infty} \left(2^{-1}\right)^{\ell} = 2^n \cdot \frac{1}{1-2^{-1}} = 2^{n+1}$$

② $n > 0$

$k = -k$ için

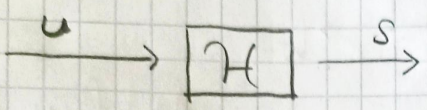
$$y(n) = \sum_{k=-\infty}^{0} 2^k = \sum_{k=0}^{\infty} \left(2^{-1}\right)^k = \frac{1}{1-2^{-1}} = 2$$

Birim dürtü tepkisi



$$\delta(t) = \frac{d\,u(t)}{dt}$$

Birim basamak tepkisi

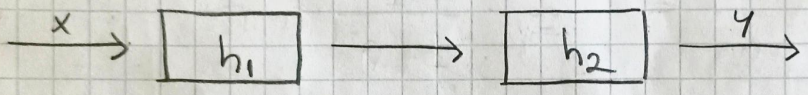


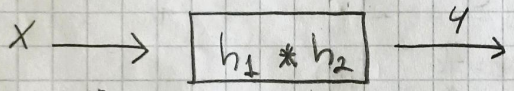
$$h(t) = \frac{d\,s(t)}{dt}$$

$$\delta(n) = u(n) - u(n-1)$$

$$h(n) = s(n) - s(n-1)$$

LTI Sistemlerin Bağlantısı





Örnek: $y(t) = \int_{-\infty}^{\infty} x(\tau)\,d\tau$

$$h(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau = F(\tau)\Big|_{\tau=0} = F(0) = 1$$

$$h(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} = u(t)$$

$\int_{-\infty}^{\infty} h(t)\,dt < \infty$ ise kararlıdır

$\int_{-\infty}^{\infty} u(t)\,dt = \int_{0}^{\infty} dt = \infty$ sistem kararsızdır

LTI Sistemlerin Öz Fonksiyonları

Her karmaşık üstel LTI sistemlerin öz fonksiyonlarıdır.

$$\mathcal{H}\{e^{st}\}(t) = \underbrace{H(s)}_{\text{öz değer}} \cdot \underbrace{e^{st}}_{\text{öz fonksiyon}}$$

$$H(s) = \int_{-\infty}^{\infty} h(t) \cdot e^{-st}\, dt$$

Örnek: $h(t) = \delta(t-1)$, $x(t) = e^t \cdot \cos \pi t$

a) $H(s) = ?$

b) $y(t) = ?$

$$x(t) = e^t \left[\frac{e^{j\pi t} + e^{-j\pi t}}{2} \right]$$

$x_0(t)$ $\qquad$ $x_1(t)$

$$x(t) = \frac{1}{2} e^{\underbrace{(1+j\pi)t}_{s_0}} + \frac{1}{2} e^{\underbrace{(1-j\pi)t}_{s_1}}$$

$$= a_0 \cdot e^{st} + a_1 \cdot e^{-st}$$

$$H(s) = \int_{-\infty}^{\infty} \delta(t-1) \cdot e^{-st}\, dt$$

$$= e^{-st}\Big|_{t=1} = e^{-s}$$

$$y(t) = H(s) \cdot x(t)$$

$$= H(s_0) \cdot x_0(t) + H(s_1) \cdot x_1(t)$$

$$= \frac{H(1+j\pi) \cdot e^{(1+j\pi)t}}{2} + \frac{H(1-j\pi) \cdot e^{(1-j\pi)t}}{2}$$

$H(1+j\pi) = e^{-(1+j\pi)}$

$H(1-j\pi) = e^{-(1-j\pi)}$

$$\Rightarrow \frac{e^{-(1+j\pi)} \cdot e^{(1+j\pi)t}}{2} + \frac{e^{-(1-j\pi)} \cdot e^{(1-j\pi)t}}{2}$$

$$= \frac{1}{2} e^{j\pi(t-1)} \cdot e^{(t-1)} + \frac{1}{2} e^{-j\pi(t-1)} \cdot e^{(t-1)}$$

$$= e^{t-1} \left[\frac{e^{j\pi(t-1)} + e^{-j\pi(t-1)}}{2} \right]$$

$$\boxed{y(t) = e^{t-1} \cdot \cos \pi(t-1)}$$

Ayrık zamanlı sistemler için: $H(z) = \sum_{k=-\infty}^{\infty} h(k) \cdot z^{-n}$

$x(n)$ → [H] → $H(z)\, x(n)$

$a \cdot z^n$ → $a \cdot H(z) \cdot z^n$

$z = a + j\beta$

Örnek: $h(n) = \delta(n-1)$, $x(n) = e^n \cdot \cos \pi(n)$

$H(z) = ?$

$y(n) = ?$

$$H(z) = \sum_{k=-\infty}^{\infty} \delta(n-1) \cdot z^{-k}$$

$$H(z) = z^{-k} \Big|_{k=1}$$

$$\boxed{H(z) = z^{-1}}$$

$$x(n) = e^n \cdot \frac{1}{2} \cdot e^{j\pi n} + e^n \cdot \frac{1}{2} \cdot e^{-j\pi n}$$

$$x(n) = \frac{1}{2} \underbrace{e^{(1+j\pi)n}}_{z_0} + \frac{1}{2} \underbrace{e^{(1-j\pi)n}}_{z_1}$$

$$y(n) = H(z) \cdot x(n) = H(z_0) \cdot x_0(n) + H(z_1) \cdot x_1(n)$$

$$= H(e^{(1+j\pi)}) \cdot \frac{1}{2} e^{(1+j\pi)n} + H(e^{(1-j\pi)}) \cdot \frac{1}{2} e^{(1-j\pi)n}$$

$$= e^{-(1+j\pi)} \cdot \frac{1}{2} \cdot e^{(1+j\pi)n} + e^{-(1-j\pi)} \cdot \frac{1}{2} e^{(1-j\pi)n}$$

$$= e^{n-1} \cdot \cos \pi(n-1)$$

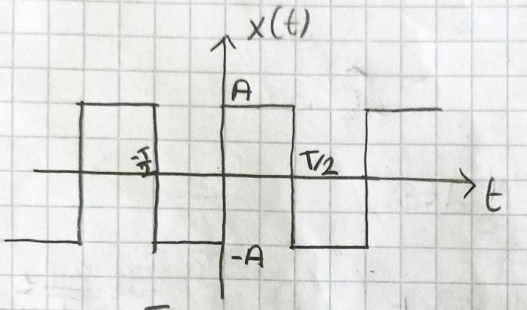
CT Fourier Serisi

$x(t) \rightarrow$ Fourier serisini açalım

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} c_k \cdot e^{jk\omega_0 t} = x(t) \qquad \begin{matrix} c \in C \\ k \in \mathbb{Z} \end{matrix}$$

$$c_k = \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t} \, dt$$

Örnek:



Fourier serisine açalım

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) \, e^{-jk\omega_0 t} \, dt$$

$\omega_0 = \frac{2\pi}{T}$

$$= \frac{1}{T} \int_{-T/2}^{0} -A \cdot e^{-jk\omega_0 t} \, dt + \frac{1}{T} \int_{0}^{T/2} A \cdot e^{-jk\omega_0 t} \, dt$$

$$= \frac{-A}{T} \frac{1}{-jk\omega_0} \cdot e^{-jk\omega_0 t} \Big|_{-T/2}^{0} + \frac{A}{T} \frac{1}{-jk\omega_0} \cdot e^{-jk\omega_0 t} \Big|_{0}^{T/2}$$

$$\left(= \frac{+A}{T} \cdot \frac{1}{jk\frac{2\pi}{T}} \cdot e^{-jk\frac{2\pi}{T} \cdot 0} - \frac{A}{\pi} \cdot \frac{1}{jk\frac{2\pi}{\pi}} \cdot e^{+jk\frac{2\pi}{\pi} \cdot \frac{+\pi}{2}} \right) +$$

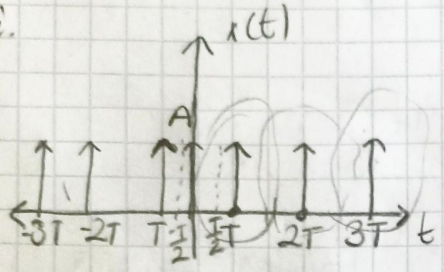
$$\left(\frac{A}{T} \cdot \frac{1}{-jk\frac{2\pi}{T}} \cdot e^{-jk\frac{2\pi}{T} \cdot \frac{T}{2}} + \left(\frac{A}{T} \cdot \frac{1}{+jk\frac{2\pi}{T}} \cdot e^{-jk\frac{2\pi}{T} \cdot 0} \right) \right)$$

$$= \frac{A}{jk2\pi} \left(1 - e^{jk\pi} \right) - \frac{A}{jk2\pi} \left(e^{-jk\pi} - 1 \right)$$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k \, e^{jk\omega_0 t} \, dt$$

$$c_k = \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t} \, dt$$

Örnek.



$$\omega_0 = \frac{2\pi}{T}$$

$$c_k = \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t} \, dt$$

$$\Rightarrow \frac{1}{T} \int_{-T/2}^{T/2} A \cdot f(t) \cdot e^{-jk\frac{2\pi}{T}t} = \frac{A}{T} \int_{-T/2}^{T/2} e^{-jk\frac{2\pi}{T}t} \, dt$$

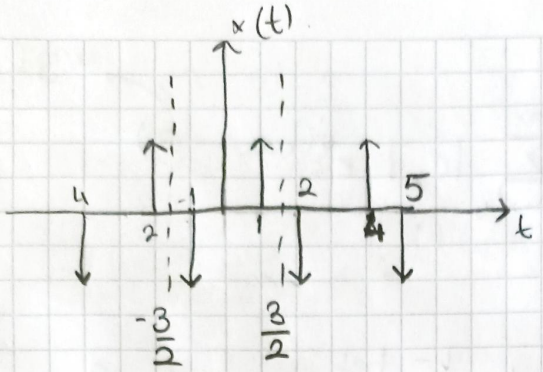
$$= \frac{A}{T} \cdot e^{-jk\frac{2\pi}{T}\cdot t} \Big|_{t=0}$$

$$= \frac{A}{T} = c_k$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{A}{T} \cdot e^{jk\frac{2\pi}{T}\cdot t} = \frac{A}{T} \sum_{k=-\infty}^{\infty} e^{jk\frac{2\pi}{T}t}$$

$$A\left(f(t) + f(t-T) + f(t-2T) + f(t-3T)\right.$$

Örnek:



$T = 3$

$$c_k = \frac{1}{T}\int_T x(t)\cdot e^{-jk\frac{2\pi}{3}t}dt$$

$$\Rightarrow \frac{1}{3}\int_{-\frac{3}{2}}^{\frac{3}{2}} x(t)\cdot e^{-jk\frac{2\pi}{3}t}dt = \frac{1}{3}\int_{-\frac{3}{2}}^{0} -\delta(t+1)\cdot e^{-jk\frac{2\pi}{3}t}dt + \frac{1}{3}\int_{0}^{\frac{3}{2}} \delta(t-1)\cdot e^{-jk\frac{2\pi}{3}t}dt$$

$$c_k = \frac{-1}{3}\cdot e^{-jk\frac{2\pi}{3}t}\Big|_{t=-1} + \frac{1}{3}\cdot e^{-jk\frac{2\pi}{3}t}\Big|_{t=1}$$

$$c_k = \frac{-1}{3}\cdot e^{jk\frac{2\pi}{3}} + \frac{1}{3}\cdot e^{-jk\frac{2\pi}{3}} = \frac{2j}{3}\cdot\left(\frac{e^{-jk\frac{2\pi}{3}} - e^{jk\frac{2\pi}{3}}}{2j}\right)$$

$$\Rightarrow c_k = \frac{2j}{3}\cdot\sin\left(k\cdot\frac{2\pi}{3}\right)$$

$$x(t) = \sum_{k=-\infty}^{\infty} -\frac{2j}{3}\cdot\sin\left(k\frac{2\pi}{3}\right)\cdot e^{jk\frac{2\pi}{3}t}$$

Gibbs Fenomeni

$\sin\frac{2\pi}{x}$ çiz.

$c_k = |c_k| \cdot e^{j\theta}$

$x(t) = \sum_{k=-\infty}^{\infty} c_k \cdot e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} |c_k| \cdot e^{j(k\omega_0 t + \theta)}$ ($\theta \to \arg(c_k)$)

Örnek

$$c_k = \begin{cases} \frac{-2jA}{k\pi}, & k \text{ tek} \\ 0, & k \text{ çift} \end{cases} \qquad c_0 = 0$$

Frekans spektrumunu çizin

$z = a + jb$

$\arg(z) = \tan^{-1}\left(\frac{b}{a}\right)$

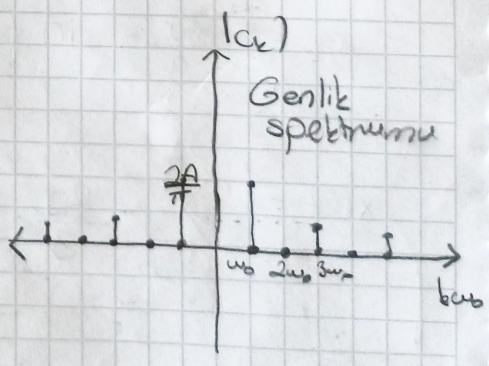
$$|c_k| = \begin{cases} \frac{2A}{|k|\pi}, & k \text{ tek} \\ 0, & k \text{ çift} \end{cases}$$

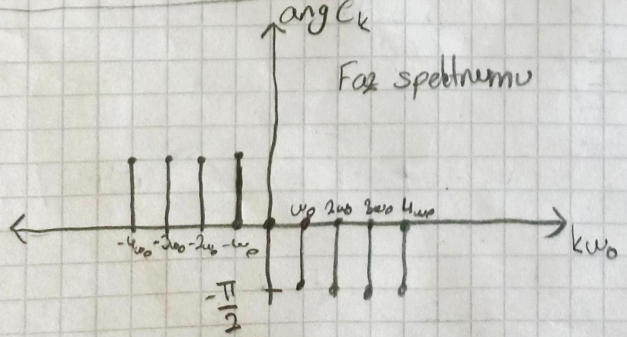
$$\arg c_k = \begin{cases} \arg\frac{-2jA}{k\pi}, & k \text{ tek} \\ \arg 0, & k \text{ çift} \end{cases}$$

$\arg 0 = 0$

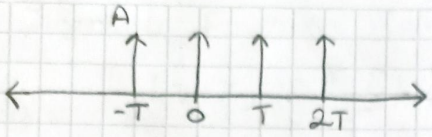
$\arg\frac{-2jA}{k\pi} = \frac{-\pi}{2}$ $k > 0$ iken

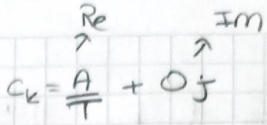
$\arg\frac{2jA}{k\pi} = \frac{\pi}{2}$ $k < 0$





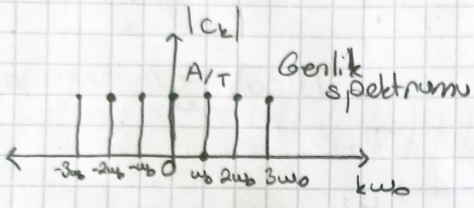
$$|c_k| = \sqrt{Re(c_k)^2 + Im(c_k)^2}$$

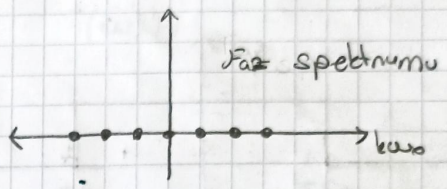




$$c_k = \frac{A}{T} + 0j$$

$$\arg c_k = \tan^{-1}\frac{0}{A/T} = 0$$

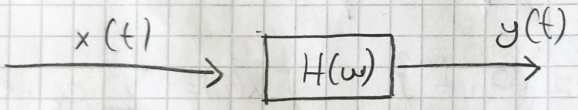




Frekans Tepkisi

Örnek $H(\omega) = e^{-j\frac{\omega}{4}}$ — $H(\omega)$: Frekans tepkisi

$x(t) = \frac{1}{2}\cos 2\pi t$ ⟶ $\omega_0 = 2\pi$



$$x(t) = \frac{1}{2}\cdot\left[\frac{e^{j2\pi t} + e^{-j2\pi t}}{2}\right] \qquad \omega_0 = 2\pi \cdot f$$

$x(t) \overset{FS}{\longleftrightarrow} c_k$ $\qquad c_1 = c_{-1} = \frac{1}{4}$ $\qquad H(-2\pi) = e^{j\frac{\pi}{2}}$ $\qquad H(2\pi) = e^{-j\frac{\pi}{2}}$

$$x(t) = \frac{1}{2}\left[\frac{e^{j\cdot(1)\cdot 2\pi t} + e^{j\cdot(-1)2\pi t}}{2}\right]$$

$$x(t) = \frac{1}{4}\cdot e^{j\cdot(1)\cdot 2\pi t} + \frac{1}{4}\cdot e^{j(-1)2\pi t}$$

$$y(t) = \sum_{k=-1}^{1} H(k2\pi)\cdot c_k e^{jk2\pi t} = H(-2\pi)\cdot c_{-1} e^{-j2\pi t} + H(2\pi)\cdot c_1 e^{j2\pi t}$$

$$= e^{j\frac{\pi}{2}}\cdot\frac{1}{4}\cdot e^{-j2\pi t} + e^{-j\frac{\pi}{2}}\cdot\frac{1}{4}\cdot e^{j2\pi t}$$

$$y(t) = \frac{1}{4}\cdot\left(e^{-j\left(2\pi t - \frac{\pi}{2}\right)} + e^{j\left(2\pi t - \frac{\pi}{2}\right)}\right) = \frac{1}{2}\cos\left(2\pi t - \frac{\pi}{2}\right)$$

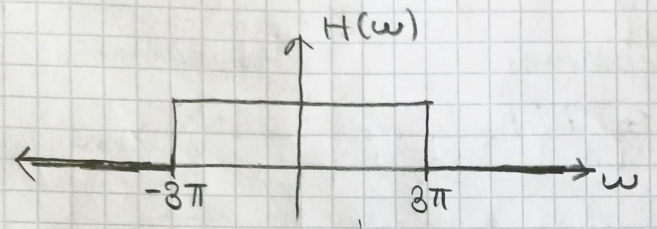
overlap
88 - 108 MHz
radyo yayını

Filtreleme 6

Örnek:

$$H(\omega) = \begin{cases} 1, & |\omega| \leq 3\pi \\ 0, & \text{diğer} \end{cases}$$

$$x(t) = 1 + 2\cos 2\pi t + \cos 4\pi t + \frac{1}{2}\cos 6\pi t$$



Alçak geçiren kısıt

$$x(t) = \underset{k=0}{\frac{1}{3}} + \underset{k=1}{e^{j\cdot 1\cdot 2\pi t}} + \underset{k=-1}{e^{j(-1)\cdot 2\pi t}} + \frac{\underset{k=2}{e^{j\cdot 2\cdot 2\pi t}} + \underset{k=-2}{e^{j\cdot(-2)\cdot 2\pi t}}}{2} + \frac{\underset{k=3}{e^{j\cdot 3\cdot 2\pi t}} + \underset{k=-3}{e^{j(-3)\cdot 2\pi t}}}{4}$$

$$c_k = \begin{cases} 1, & k \in \{-1, 0, 1\} \\ 1/2, & k \in \{-2, 2\} \\ 1/4, & k \in \{-3, 3\} \\ 0, & \text{diğer} \end{cases}$$

$$x(t) = \sum_{k=-3}^{3} c_k \cdot e^{jk2\pi t}$$

$H(-6\pi) = 0$ $\quad H(2\pi) = 1$
$H(6\pi) = 0$ $\quad H(-2\pi) = 1$
$H(-4\pi) = 0$ $\quad H(0) = 1$
$H(4\pi) = 0$

$$y(t) = \sum_{k=-3}^{3} \underbrace{H(k2\pi)\cdot c_k}_{b_k} \cdot e^{jk2\pi t} =$$

$b_{-3} = H(-6\pi)\cdot c_{-3} = 0\cdot\frac{1}{4} = 0$ $\quad b_3 = 0$

$b_{-2} = H(-4\pi)\cdot c_{-2} = 0\cdot\frac{1}{2} = 0$ $\quad b_2 = 0$

$b_{-1} = H(-2\pi)\cdot c_{-1} = 1\cdot 1 = 1$ $\quad b_1 = 1$

$b_0 = H(0)\cdot c_0 = 1\cdot 1 = 1$

$$b_k = \begin{cases} 1, & k \in (-1, 0, 1) \\ 0, & \text{diğer} \end{cases}$$

$$y(t) = \sum_{k=-1}^{1} b_k \cdot e^{jk2\pi t} = b_{-1}\cdot e^{-j2\pi t} + b_0 e^{j0\cdot 2\pi t} + b_1 \cdot e^{j2\pi t}$$

$$= e^{-j2\pi t} + 1 + e^{j2\pi t} = 1 + 2\cdot\cos 2\pi t$$

Ayrık Zamanlı Fourier Serisi

$$x(n) = \sum_{k=\langle N \rangle} c_k \cdot e^{jk\Omega_0 n} \qquad \Omega_0 = \frac{2\pi}{N}$$

$$c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) \cdot e^{-j\left(\frac{2\pi}{N}\right)n}$$

Örnek: $x(n) = \sin\left(\left(\frac{2\pi}{7}\right)n\right)$

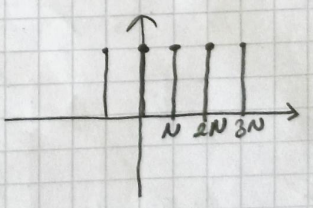
$$N = \frac{2\pi}{\frac{2\pi}{7}} = 7$$

$$x(n) = \frac{1}{2j} \cdot e^{j1\cdot\Omega_0 n} - \frac{1}{2j} e^{j\cdot(-1)\Omega_0 n}$$

$$c_k = \begin{cases} -j/2 & , k = 1 \\ j/2 & , k = -1 \\ 0 & , k \in \{-3, -2, 0, 2, 3\} \end{cases}$$

$$x(n) = \sum_{k=-3}^{3} c_k \cdot e^{jk\frac{2\pi}{7}n}$$

$$x(n) = \sum_{l=-\infty}^{\infty} \delta(n - N \cdot l)$$



Örnek: $x(n) = \begin{cases} -1/2 , & n=-1 \\ 1 , & n=0 \\ 1/2 , & n=1 \\ 0 , & n \in \{-2, 2\} \end{cases}$

$N = 5$

$$c_k = \frac{1}{5} \sum_{n=-2}^{2} x(n) \cdot e^{-jk\frac{2\pi}{5}n}$$

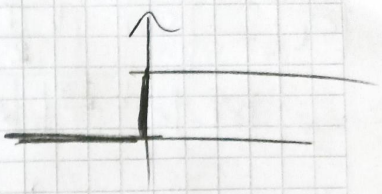
$$\frac{1}{5}\left(\frac{-1}{2} \cdot e^{jk\frac{2\pi}{5}} + 1 \cdot e^{-jk\frac{2\pi}{5}\cdot 0} + \frac{1}{2} \cdot e^{-jk\frac{2\pi}{5}} \right)$$

$$\frac{-1 \cdot 2j}{2}\left(\frac{e^{+jk\frac{2\pi}{5}} + e^{-jk\frac{2\pi}{5}}}{2j} \right) + 1$$

$$c_k = \frac{1}{5} \cdot \left(1 - j\sin\left(k\frac{2\pi}{5}\right) \right)$$

Soru: $y(n) = 2x(n) - x(n-1)$

a) Birim dürtü tepkisi nedir? $h(n)$



$x(n) = \delta(n)$

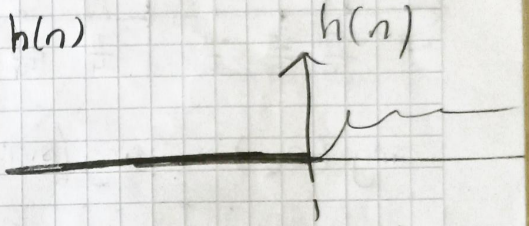
$h(n) = 2 \cdot \delta(n) - \delta(n-1)$

$\delta(n-1) \longrightarrow [H] \longrightarrow 2^n \cdot u(n-1)$

Sistemin birim dürtü tepkisi?

$n = n+1$ için

$\delta(n) \longrightarrow [H] \longrightarrow 2^{n+1} \cdot u(n)$

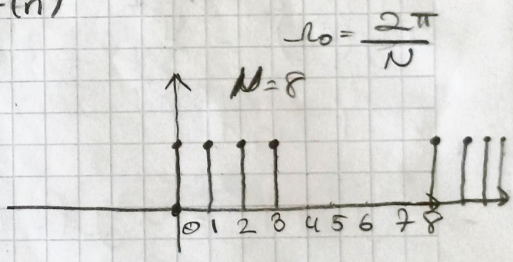


Bir sistemin nedensel olabilmesi için $h(n) = 0$, $n < 0$ iken

Bir sistem belleksiz ise $h(n) = k \cdot \delta(n)$

Örnek

$$x(n) = \begin{cases} 1, & n \in [0..3] \\ 0, & n \in [4..7] \end{cases}$$



$\Omega_0 = \frac{2\pi}{N}$

$x(n) \overset{FS}{\longleftrightarrow} c_k$ $\qquad c_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-jk\Omega_0 n}$

$\Omega_0 = \frac{\pi}{4}$

$c_k = \frac{1}{N} \sum_{n=0}^{7} x(n) \cdot e^{-jk\frac{\pi}{4}n}$ $\qquad c_k = \frac{1}{8} \sum_{n=0}^{3} e^{-jk\frac{\pi}{4}n}$ $\qquad \alpha \quad \Omega = \frac{2\pi}{8} = \frac{\pi}{4}$

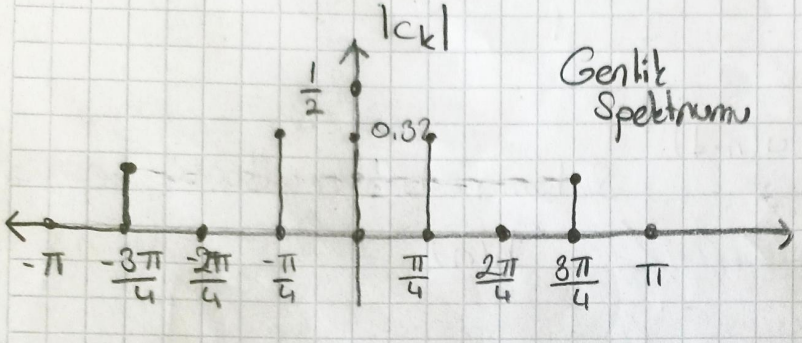
Tanımsızlık durumunda analiz denkleminde "0" yaz.

Sonlu toplam formülü:

$$c_k = \frac{1}{8} \cdot \frac{1 - e^{\left(-jk\frac{\pi}{4}\right)^4}}{1 - e^{-jk\frac{\pi}{4}}} = \frac{1}{8 \cdot e^{jk\frac{3\pi}{8}}} \cdot \frac{\sin k\frac{\pi}{2}}{\sin k\frac{\pi}{8}}$$

$$c_k = \frac{1}{8} \frac{e^{-jk\frac{\pi}{2}} \left(e^{jk\frac{\pi}{2}} - e^{-jk\frac{\pi}{2}}\right)}{e^{-jk\frac{\pi}{8}} \left(e^{jk\frac{\pi}{8}} - e^{-jk\frac{\pi}{8}}\right)}$$

$$|c_k| = \frac{\left|\sin\left(k\frac{\pi}{2}\right)\right|}{|8| \cdot \left|e^{-jk\frac{3\pi}{8}}\right| \left|\sin\left(k\frac{\pi}{8}\right)\right|} = \frac{\left|\sin\left(k\frac{\pi}{2}\right)\right|}{8 \cdot \left|\sin\left(k\frac{\pi}{8}\right)\right|}$$



$$\boxed{\arg(a+jb) = \tan^{-1}\frac{b}{a}}$$

$$\arg(c_k) = \arg\left(\sin k\frac{\pi}{2}\right) - \cancel{\arg(8)}^{0} - \arg e^{-jk\frac{3\pi}{8}} - \arg\sin\frac{k\pi}{8}$$

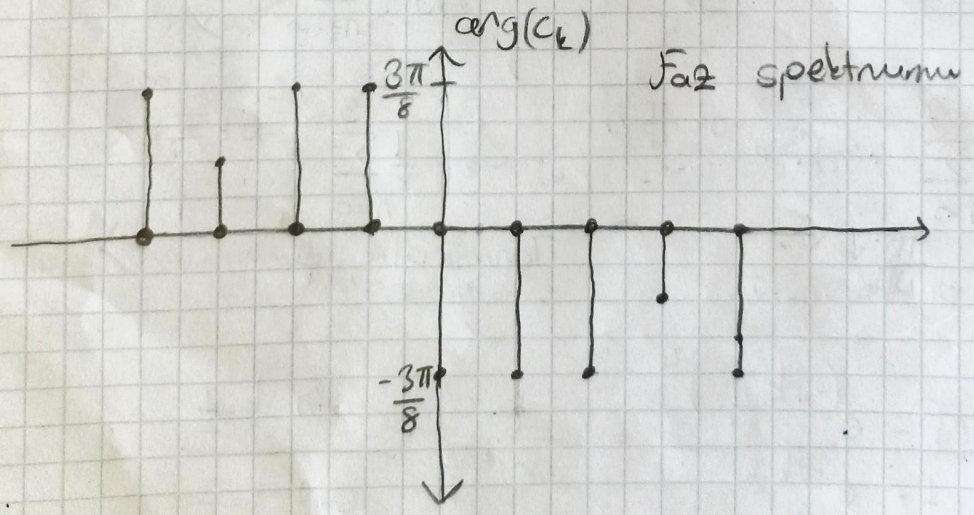
$$\arg(c_1) = \arg\left(\sin\left(\frac{\pi}{2}\right)\right) - \frac{3\pi}{8} - \arg\left(\sin\left(\frac{3\pi}{8}\right)\right) \quad k\frac{3\pi}{8}$$

$$= -\frac{3\pi}{8}$$

$$\arg(c_0) = \arg\left(\frac{1}{2}\right) = 0$$

$k=-1$ için

$$\arg(c_{-1}) = \underbrace{\arg\left(\sin\left(-\frac{\pi}{2}\right)\right)}_{\pi} + \frac{3\pi}{8} - \underbrace{\arg\sin\frac{-\pi}{8}}_{\pi} = \frac{3\pi}{8}$$



$T_2 = 2$

$4 \cdot T_2 = 2T_1$ ⇒ EKOK(4,2) = 4

Örnek: LTI → $h(n) = \left(\frac{1}{2}\right)^{n+1} \cdot u(n)$

$$x(n) = 1 + \frac{1}{2}\cos\left(\frac{\pi}{2}n\right) + \frac{1}{8}\cos(\pi n)$$ y(n) nedir?

euler açılım

$x(n) \rightarrow [H] \rightarrow y(n)$

$$x(n) = 1 \cdot e^{j0} + \frac{1}{4}e^{j\frac{\pi}{2}n} + \frac{1}{4}e^{-j\frac{\pi}{2}n} + \frac{1}{8}e^{j\pi n}$$

$N = 4$ $\quad k \in \{-1, 0, 1, 2\}$ → $c_k \; e^{j\pi n}$

$$x(n) = \sum_{k=-1}^{2} c_k \cdot e^{jk\frac{2\pi}{4}n}$$?

$\frac{2\pi}{N}$

$e^{jk\Omega n}$ $\quad \Omega = \frac{2\pi}{N}$ $\quad z = e^{j\Omega}$ $\quad e^{j\frac{\pi}{2}n}$

giriş karmaşık bir üstel ise

$$y(n) = \sum c_k H\left(k\frac{2\pi}{N}\right) \cdot e^{jk\frac{2\pi}{N}n}$$ Çıkış bu şekilde hesaplanır

$$H(\Omega) = \sum h(n) \cdot e^{-j\Omega n} \Rightarrow \sum_{n=0}^{\infty} \frac{1}{2} \cdot \left(\frac{1}{2} \cdot e^{-j\Omega}\right)^n = \frac{1}{2}\sum_{n=0}^{\infty} \alpha^n$$

$\alpha = \frac{1}{2}e^{-j\Omega}$

$$= \frac{1}{1 - \frac{1}{2}e^{-j\Omega}} = \frac{1}{2 - e^{-j\Omega}}$$

$c_k \cdot H\left(k\frac{2\pi}{N}\right) = b_k$ için

$y(n) \overset{FD}{\longleftrightarrow} b_k$

$b_{-1} = c_{-1} \cdot H\left(\frac{-\pi}{2}\right) \Rightarrow H\left(\frac{-\pi}{2}\right) = \frac{1}{2 - e^{j\frac{\pi}{2}}} = \frac{1}{2-j} \cdot \left(\frac{1}{4}\right)$

$b_0 = c_0 \cdot H(0) = c_0 = \frac{1}{2}$

$b_1 = c_1 \cdot H\left(\frac{\pi}{2}\right) = \frac{1}{4} \cdot \frac{1}{1+2j} = \frac{1}{8+4j}$

$b_2 = c_2 H(\pi) = \frac{1}{8} \cdot \frac{1}{3} = \frac{1}{24}$

b_k genlik ve faz spektrumu

$$y(n) = \frac{e^{-j\frac{\pi}{2}n}}{8-4j} + \frac{1}{2} + \frac{e^{j\frac{\pi}{2}n}}{8+4j} + \frac{e^{j\pi n}}{24}$$

Örnek: $H(\Omega) = e^{-j\Omega}$

$$x(n) = \frac{1}{2}\cos\frac{2\pi}{5}n$$

$$x(n) = \frac{1}{4}e^{j\frac{2\pi}{5}n} + \frac{1}{4}e^{-j\frac{2\pi}{5}n}$$

$$c_k = \begin{cases} \frac{1}{4}, & k \in \{-1, 1\} \\ 0, & k \in \{-2, 0, 2\} \end{cases}$$

$$y(n) = \sum_{k=-1}^{1} \underbrace{c_k \cdot H\left(k \cdot \frac{2\pi}{5}\right)}_{b_k} \cdot e^{jk\frac{2\pi}{5}n}$$

$$b_1 = c_1 \cdot H\left(\frac{2\pi}{5}\right) = \frac{1}{4} \cdot e^{-j\frac{2\pi}{5}}$$

$$b_{-1} = c_{-1} H\left(-\frac{2\pi}{5}\right) = \frac{1}{4}e^{j\frac{2\pi}{5}}$$

$$y(n) = b_{-1} \cdot e^{-j\frac{2\pi}{5}n} + b_1 \cdot e^{j\frac{2\pi}{5}n}$$

$$y(n) = \frac{1}{4} \cdot e^{j\frac{2\pi}{5}} \cdot e^{-j\frac{2\pi}{5}} + \frac{1}{4} \cdot e^{j\frac{2\pi}{5}} \cdot e^{-j\frac{2\pi}{5}n}$$

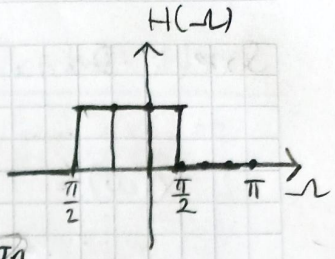
$$y(n) = \frac{1}{4} \cdot e^{-j\frac{2\pi}{5}(n-1)} + \frac{1}{4} \cdot e^{j\frac{2\pi}{5}(n-1)} = \frac{1}{2}\cos\left(\frac{2\pi}{5}(n-1)\right)$$

$$j\frac{2\pi}{5} + \left(-j\frac{2\pi}{5}n\right|$$

$$-j\frac{2\pi}{5}(n-1)$$

Örnek:

$$H(\Omega) = \begin{cases} 1, & |\Omega| \le \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < |\Omega| \le \pi \end{cases}$$



$$x(n) = 1 + \cos\frac{\pi}{4}n + \frac{1}{2}\cos\frac{3\pi}{4}n + \frac{1}{5}\cos\pi n$$

Alçak geçiren bant

$$x(n) = 1 + \left(\frac{1}{2}\cdot e^{j\frac{2\pi}{8}n} + \frac{1}{2}e^{-j\frac{2\pi}{8}n}\right) + \left(\frac{1}{4}e^{j3\cdot\frac{2\pi}{8}n} + \frac{1}{4}e^{-j3\frac{2\pi}{8}n}\right) + \frac{1}{5}\cdot e^{j4\frac{2\pi}{8}n}$$

$$x(n) \xleftrightarrow{FS} c_k = \begin{cases} \frac{1}{4}, & k \in \{-3, 3\} \\ \frac{1}{2}, & k \in \{-1, 1\} \\ 1, & k = 0 \\ \frac{1}{5}, & k = 4 \\ 0, & k = \{-2, 2\} \end{cases}$$

$$b_k = c_k \cdot H\left(k\cdot\frac{2\pi}{8}\right)$$

$b_3 = b_{-3} = 0$

$b_2 = b_{-2} = 0$

$b_1 = c_1 \cdot H\left(\frac{\pi}{4}\right) = \frac{1}{2}\cdot 1 = \frac{1}{2}$

$b_{-1} = \frac{1}{2}\cdot 1 = \frac{1}{2}$

$b_4 = \frac{1}{5}\cdot 0 = 0$

$b_0 = c_0 \cdot H(0) = 1\cdot 1 = 1$

$$b_k = \begin{cases} 1, & k = 0 \\ 1/2, & k \in \{-1, 1\} \\ 0, & k \in \{-3, -2, 2, 3, 4\} \end{cases}$$

$$y(n) = \sum_{k=-1}^{1} b_k \cdot e^{jk\frac{\pi}{4}n} = \frac{1}{2}\cdot e^{-j\frac{\pi}{4}n} + 1\cdot 1 + \frac{1}{2}\cdot e^{j\frac{\pi}{4}n}$$

$$1 + \frac{1}{2}\left(\frac{e^{j\frac{\pi}{4}n} + e^{-j\frac{\pi}{4}n}}{2}\right)$$

$$\cos\frac{\pi}{4}n$$

Sürekli Zamanlı Fourier Dönüşümü

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt = \mathcal{F}\{x(t)\}$$

$$\mathcal{F}^{-1}\{X(\omega)\} = x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t}\, d\omega$$

Örnek:

$x(t) = A \cdot \delta(t - t_0)$ $\qquad x(t) \overset{FT}{\longleftrightarrow} X(\omega)$

$X(\omega) = ?$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt \Rightarrow A \int_{-\infty}^{\infty} \delta(t - t_0) \cdot e^{-j\omega t}\, dt$$

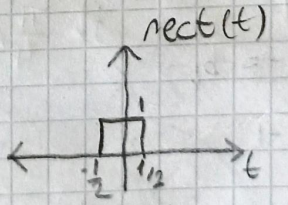
$$= A \cdot \int_{-\infty}^{\infty} e^{-j\omega t}\, dt = A \cdot e^{-j\omega t} \Big|_{t=t_0} = A \cdot e^{-j\omega t_0}$$

Örnek: $X(\omega) = 2\pi \cdot \delta(\omega - \omega_0)$, $x(t) = ?$

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \cdot \delta(\omega - \omega_0)\, e^{j\omega t}\, d\omega = \int_{-\infty}^{\infty} \delta(\omega - \omega_0) \cdot e^{j\omega t}\, d\omega$$

$$= e^{j\omega t} \Big|_{\omega = \omega_0} = e^{j\omega_0 t}$$

Örnek: $x(t) = \text{rect}(t)$



$X(\omega) = ?$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt$$

$$= \int_{-1/2}^{1/2} e^{-j\omega t}\, dt = \frac{1}{j\omega} e^{-j\omega t} \Big|_{-1/2}^{1/2} = \frac{1}{j\omega}\left(e^{-\frac{j\omega}{2}} + e^{\frac{j\omega}{2}}\right)$$

$$= \frac{\sin\left(\frac{\omega}{2}\right)}{\frac{\omega}{2}} = \text{sinc}\left(\frac{\omega}{2}\right)$$

Örnek: $x(t) = A.\cos(\omega_0 t + \theta)$, $X(\omega) = ?$

$$q(t) = A.\cos(\omega_0 t) \xleftrightarrow{FT} \pi.\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]$$

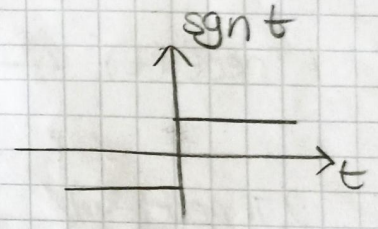
$$x(t) = q\left(t + \frac{\theta}{\omega_0}\right)$$

$$X(\omega) = e^{j\omega\left(\frac{\theta}{\omega_0}\right)} \quad Q(\omega) = e^{j\omega\left(\frac{\theta}{\omega_0}\right)}.\ \pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$$

Örnek: $x(t) = 3.\cos 5t \xleftrightarrow{FT} X(\omega)$

$$X(\omega) = 3.\ \pi\left[\delta(\omega-5) + \delta(\omega+5)\right]$$

Örnek: $x(t) = u(t)$, $X(\omega) = ?$



$$u(t) = \frac{\operatorname{sgn} t}{2} + \frac{1}{2}$$

$$X(\omega) = \frac{1}{2}\mathcal{F}\{\operatorname{sgn} t\} + \frac{1}{2}\mathcal{F}\{1\}$$

$$\frac{1}{2}\cdot\frac{2}{j\omega} + \frac{1}{2}\cdot 2\pi\delta(\omega) = \frac{1}{j\omega} + \delta(\omega).\pi$$

$e^{j\omega_0 t}.x(t) \xleftrightarrow{CTFT} X(\omega-\omega_0)$ Frekansta kaydırma

Örnek: $x(t) = \cos\omega_0 t.\cos 20\pi t$ $X(\omega) = ?$

$$q(t) = \cos\omega_0 t \Rightarrow Q(\omega) = \pi.\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]$$

$$x(t) = q(t).\cos 20\pi t$$

$$x(t) = \underbrace{\frac{1}{2}e^{j20\pi t}.q(t)}_{\frac{1}{2}Q(\omega-20\pi)} + \underbrace{\frac{1}{2}e^{-j20\pi t}.q(t)}_{\frac{1}{2}Q(\omega+20\pi)}$$

$$\frac{1}{2}\pi\left(\delta(\omega-20\pi-\omega_0)+\delta(\omega-20\pi+\omega_0)\right) + \frac{\pi}{2}\left[\delta(\omega+20\pi-\omega_0)+\delta(\omega+20\pi+\omega_0)\right]$$

- $x(at) \xleftrightarrow{CTFT} \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$

Örnek: $rect(at) = x(t)$ $\quad X(\omega) = ?$

$x(t) = rect(t) \xleftrightarrow{CTFT} X(\omega) = sinc\left(\frac{\omega}{2}\right)$

$x(at) \xleftrightarrow{CTFT} \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \Rightarrow X(\omega) = \frac{1}{|a|} sinc\left(\frac{\omega}{2a}\right)$

- $X(t) \xleftrightarrow{CTFT} 2\pi\, x(-\omega)$ Dualite özelliği

Örnek: $x(t) = sinc\left(\frac{t}{2}\right) \xleftrightarrow{CTFT} X(\omega)$

$rect(t) \longleftrightarrow sinc\left(\frac{\omega}{2}\right)$ $\quad sinc\left(\frac{t}{2}\right) \xleftrightarrow{CTFT} 2\pi\, rect(\omega)$

$2\pi\, rect(-\omega) \longleftrightarrow sinc\left(\frac{t}{2}\right)$

$X(\omega) = 2\pi \cdot rect(\omega)$

- $x_1(t) \xleftrightarrow{CTFT} X_1(\omega)$ $\quad x_2(t) \xleftrightarrow{CTFT} X_2(\omega)$

$x_1 * x_2(t) \xleftrightarrow{CTFT} X_1(\omega) \cdot X_2(\omega)$

Örnek: $x(t) = e^{-2t} \cdot u(t)$

$h(t) = u(t)$ $\quad y(t) = x * h(t)$

$X(\omega) = \frac{1}{2 + j\omega}$

$H(\omega) = \pi \cdot \delta(\omega) + \frac{1}{j\omega}$

$y(\omega) = \frac{\pi}{2}\delta(\omega) + \frac{1}{j2\omega - \omega^2}$

c: cannier

Örnek: $y(t) = x(t) \cdot \cos(\omega_c t)$ $\quad Y(\omega) = ?$

$u(t) = \cos(\omega_c t)$ $\quad U(\omega) = \pi \cdot \left[\delta(\omega - \omega_c) + \delta(\omega + \omega_c) \right]$

$$Y(\omega) = \frac{1}{2\pi} X(\omega) * U(\omega)$$

$$2\pi\, Y(\omega) = \int_{-\infty}^{\infty} X(\lambda) \cdot U(\quad)\, d\omega = \int$$

Örnek: $x(t) = \frac{d}{dt} f(t)$ $\quad X(\omega) = ?$ Zamanda türev alma

$g(t) = f(t) \qquad G(\omega) = 1$

$x(t) = \frac{d}{dt} g(t) \longleftrightarrow j\omega\, G(\omega) = j\omega$

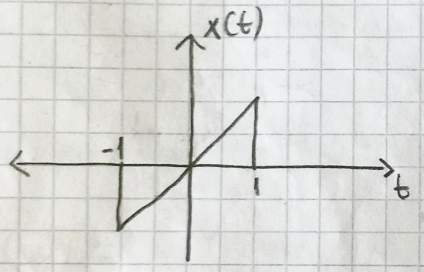
Örnek: $x(t) = t \cdot \cos\omega_0 t$, $X(\omega) = ?$ Frekansta türev alma

$\cos\omega_0 t = g(t) \Rightarrow G(\omega) = \pi \cdot [\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$

$x(t) = t \cdot g(t)$

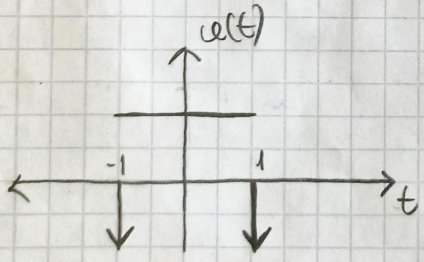
$X(\omega) = j \frac{d\, G(\omega)}{d\omega}$

Örnek:



Zamanda integral alma

$g(t) = \frac{d\, x(t)}{dt}$



$g(t) = -\delta(t-1) - \delta(t+1) + \text{rect}\left(\frac{t}{2}\right)$

$G(\omega) = -e^{-j\omega} - e^{j\omega} + 2 \cdot \text{sinc}\,\omega$

$x(t) = \int g(t)\, dt$

$X(\omega) = \frac{G(\omega)}{j\omega} + \pi \cdot G(0) \cdot \delta(\omega)$

→ özellik

Zamanda integral alma

Örnek: $x(t) = u(t) \longleftrightarrow X(\omega) = ?$

$$u(t) = \int_0^t \delta(\tau)\, d\tau$$

$$\int f(t)\, dt \overset{CTFT}{\longleftrightarrow} \frac{1}{j\omega} + \pi \cdot \delta(0) \cdot \delta(\omega)$$

Örnek: $x(t) = \text{sinc}\left(\frac{t}{2}\right)$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} \left|\text{sinc}\left(\frac{t}{2}\right)\right|^2 dt$$

$$E_x = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$\text{rect}(t) \longleftrightarrow \text{sinc}\left(\frac{\omega}{2}\right)$$

$$2\pi\, \text{rect}(\omega) \longleftrightarrow \text{sinc}\left(\frac{t}{2}\right)$$

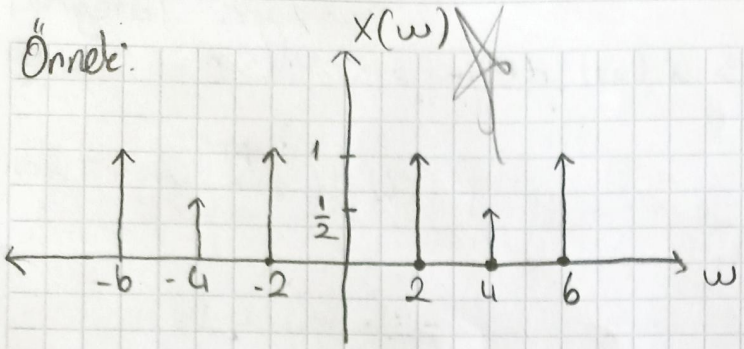
$X(\omega)$

$$E_x = \frac{1}{2\pi} \int_{-\infty}^{\infty} |2\pi \cdot \text{rect}(\omega)|^2 d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} 4\pi^2 \cdot \text{rect}^2(\omega)\, d\omega$$

$$= \frac{4\pi^2}{2\pi} \int_{-\frac{1}{2}}^{\frac{1}{2}} d\omega = 2\pi \left(\omega \Big|_{-\frac{1}{2}}^{\frac{1}{2}} \right) = 2\pi \text{ joule}$$

Örnek:

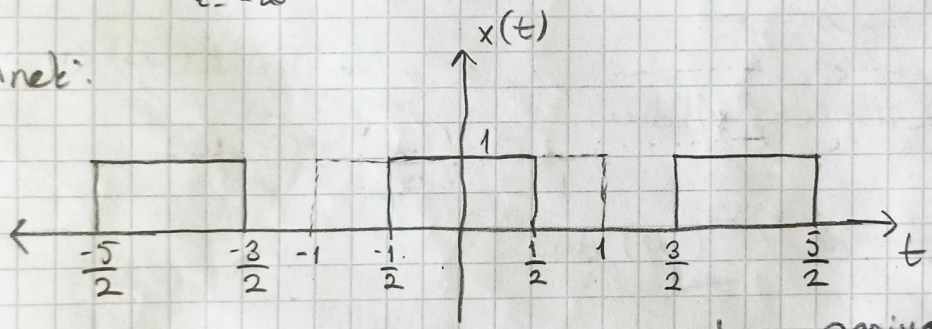


x(t) periyodik midir?

$$X(\omega) = \delta(\omega-2) + \delta(\omega+2) + \frac{1}{2}\delta(\omega-4) + \frac{1}{2}\delta(\omega+4) + \delta(\omega-6) + \delta(\omega+6)$$

$$X(\omega) = \sum_{k=-\infty}^{\infty} a_k \cdot \delta(\omega - k\omega_0) \qquad \omega_0 = 2 \qquad T = \frac{2\pi}{\omega_0} = \pi$$

Örnek:



$T = 2$

$\omega_0 = \frac{2\pi}{2} = \pi$

→ bir periyot için

$$x_T(t) \xleftrightarrow{FT} \text{sinc}\left(\frac{\omega}{2}\right) = X_T(\omega)$$

$$a_k = \frac{1}{T} \cdot X_T(k\omega_0) = \frac{1}{2} X_T(k\pi) = \frac{1}{2} \text{sinc}\left(\frac{k\pi}{2}\right) = a_k$$

$$X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi \cdot a_k \delta(\omega - k\omega_0) = \sum_{k=-\infty}^{\infty} \pi \cdot \text{sinc}\left(\frac{k\pi}{2}\right) \delta(\omega - k\pi)$$

Örnek: $x(t) \xleftrightarrow{FT} X(\omega)$ $\quad y(t) = \frac{d^2 x(t-2)}{dt^2}$

$y(t) \xleftrightarrow{FT} Y(\omega)$

$Y(\omega) = ?$

$v(t) = x(t-2) \xleftrightarrow{FT} V(\omega) = e^{-j2\omega} \cdot X(\omega)$

$y(t) = \frac{d^2}{dt^2} v(t) \xleftrightarrow{FT} Y(\omega) = (j\omega)^2 \cdot V(\omega)$

$= -\omega^2 \cdot e^{-j2\omega} \cdot X(\omega)$

Örnek: $y(t) = x(at-b)$, $Y(\omega) = ?$

$v(t) = x(t-b) \xleftrightarrow{FT} V(\omega) = e^{-j\omega b} \cdot X(\omega)$

$v(at) = x(at-b) \xleftrightarrow{FT} \frac{1}{|a|} V\left(\frac{\omega}{a}\right) = Y(\omega)$

$Y(\omega) = \frac{1}{|a|} \cdot e^{-j\omega \frac{b}{a}} \cdot X\left(\frac{\omega}{a}\right)$

Örnek: $\sum_{k=-\infty}^{\infty} x_0(t - kT)$ $\quad X(\omega) = ?$

$x_0(t) = A \cdot \text{rect}\left(\frac{2t}{T}\right) = A \cdot \text{rect}\left(\frac{t}{\frac{T}{2}}\right)$

$x_T(t) = x_0(t) = A \cdot \text{rect}\ \frac{2t}{T}$

$\text{rect}(t) \xleftrightarrow{FT} \text{sinc}\left(\frac{\omega}{2}\right)$

$\text{rect}(at) \xleftrightarrow{FT} \frac{1}{|a|} \text{sinc}\left(\frac{\omega}{2a}\right)$

$A \cdot \text{rect}\left(\frac{2}{T} \cdot t\right) \longleftrightarrow \frac{A \cdot T}{2} \text{sinc}\left(\frac{T \cdot \omega}{4}\right) = X_T(\omega)$

$a_k = \frac{1}{T} X_T(k\omega_0) = \frac{1}{T} \cdot \frac{AT}{2} \cdot \text{sinc}\left(\frac{T \cdot k \frac{2\pi}{T}}{4}\right)$

$a_k = \frac{A}{2} \text{sinc}\ \frac{k\pi}{2}$

Örnek: $x(t) = \int_{-\infty}^{t} e^{-(3+j2)\tau}\, u(\tau)\, d\tau$ $\qquad X(\omega) = ?$

$$u(t) = e^{-3t} \cdot e^{-j2t}\, u(t)$$

$$u_1(t) = e^{-3t} \cdot u(t) \longleftrightarrow U_1(\omega) = \frac{1}{3+j\omega}$$

$$u(t) = e^{-j2t} \cdot u_1(t) \longleftrightarrow U_1(\omega+2) = U(\omega)$$

$$U(\omega) = \frac{1}{3+j(\omega+2)}$$

$$x(t) = \int_{-\infty}^{t} u(\tau)\, d\tau$$

$$X(\omega) = \frac{U(\omega)}{j\omega} + \pi \cdot U(0) \cdot \delta(\omega) \quad \text{zamanda integral alma}$$

$$U(0) = \frac{1}{3+j2}$$

$$X(\omega) = \frac{1}{j\omega(3+j\omega+j2)} + \frac{\pi \cdot \delta(\omega)}{3+j2}$$

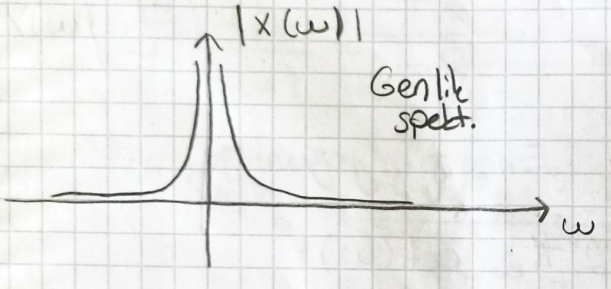
Örnek: $x(t) = \frac{1}{2}\,\text{sgn}(t-1)$, $X(\omega) = ?$

$$\text{sgn}(t) \overset{FT}{\longleftrightarrow} \frac{2}{j\omega}$$

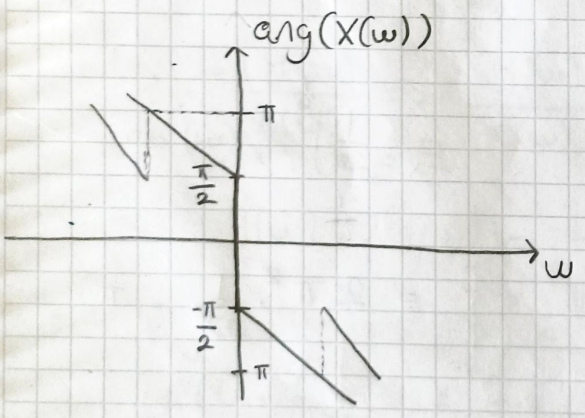
$$\frac{1}{2}\,\text{sgn}(t-1) \longleftrightarrow e^{-j\omega} \cdot \frac{2}{j\omega} \cdot \frac{1}{2}$$

$$X(\omega) = \frac{e^{-j\omega}}{j\omega}$$

$X(\omega)$ karmaşık olduğu için faz ve genlik spektrumu ayrı çizilir.



$$|X(\omega)| = \frac{|e^{-j\omega}|}{|j\omega|} = \frac{1}{\sqrt{\omega^2}} = \frac{1}{|\omega|}$$



$$\arg X(\omega) = \arg(e^{-j\omega}) - \arg(j\omega)$$

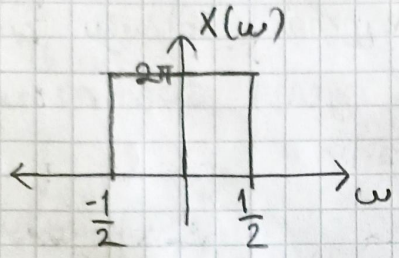
$$= \begin{cases} -\omega - \frac{\pi}{2}, & \omega > 0 \\ -\omega + \frac{\pi}{2}, & \omega < 0 \end{cases}$$

Örnek: $x(t) = \text{sinc}\left(\frac{t}{2}\right)$, $X(\omega) = ?$

$$\text{rect}(t) \longleftrightarrow \text{sinc}\left(\frac{\omega}{2}\right)$$

$$2\pi\,\text{rect}(\omega) \longleftrightarrow \text{sinc}\left(\frac{2t}{2}\right)$$

$$X(\omega) = 2\pi \cdot \text{rect}(\omega)$$



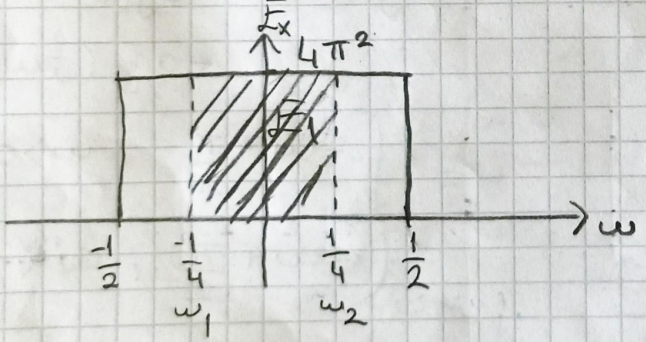
$$\frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$\frac{2\pi}{2} \cdot \text{rect}\left(\frac{\omega}{2}\right)$$

Örnek: $x(t) = \text{sinc}\left(\frac{t}{2}\right)$ Enerji yoğunluğu spektrumu?

$$E_x = |X(\omega)|^2 = |2\pi \cdot \text{rect}(\omega)|^2$$

$$E_x = 4\pi^2 \cdot \text{rect}(\omega)$$



$$E_x = \frac{1}{2\pi}\int_{-\frac{1}{4}}^{\frac{1}{4}} E_x(\omega)\,d\omega$$

$$= 2\pi^2 \cdot \frac{1}{2\pi} = \pi$$

$$E_{\text{Toplam}} = \frac{1}{2\pi}\int_{-\frac{1}{2}}^{\frac{1}{2}} 4\pi^2\,d\omega = 2\pi$$

$$\text{sinc}\,\frac{t}{2} \longleftrightarrow 2\pi\,\text{rect}\,\omega$$

$$\text{rect}(t) \longleftrightarrow \text{sinc}\,\frac{\omega}{2}$$

Örnek: $7y''(t) + 11y'(t) + 13y(t) = 5x'(t) + 3x(t)$

$y(t) \xleftrightarrow{FT} Y(\omega)$

$x(t) \xleftrightarrow{FT} X(\omega)$

$7(j\omega)^2 \cdot Y(\omega) + 11 \cdot (j\omega) \cdot Y(\omega) + 13 Y(\omega) = 5 \cdot (j\omega) X(\omega) + 3X(\omega)$

$Y(\omega)\left(-7\omega^2 + 11j\omega + 13\right) = X(\omega)\left(5j\omega + 3\right)$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{5j\omega + 3}{-7\omega^2 + 11j\omega + 13}$$

Genlik ve faz spektrumu

Örnek: $H(\omega) = \dfrac{-7\omega^2 + 11j\omega + 3}{-5\omega^2 + 2}$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \Rightarrow \frac{Y(\omega)}{X(\omega)} = \frac{-7\omega^2 + 11j\omega + 3}{-5\omega^2 + 2}$$

$-5\omega^2 Y(\omega) + 2Y(\omega) = -7\omega^2 X(\omega) + 11j\omega X(\omega) + 3X(\omega)$

$= 5(j\omega)^2 \cdot Y(\omega) + 2Y(\omega) = 7(j\omega)^2 \cdot X(\omega) + 11 \cdot j\omega X(\omega) + 3X(\omega)$

$5 \cdot y''(t) + 2 \cdot y(t) = 7 \cdot x''(t) + 11x'(t) + 3x(t)$

Önnek: $h(t) = 300 \cdot \text{sinc}(300\pi t)$

$$x(t) = \frac{1}{2} + \frac{3}{4}\cos(200\pi t) + \frac{1}{2}\cos(400\pi t) - \frac{1}{4}\cos(600\pi t)$$

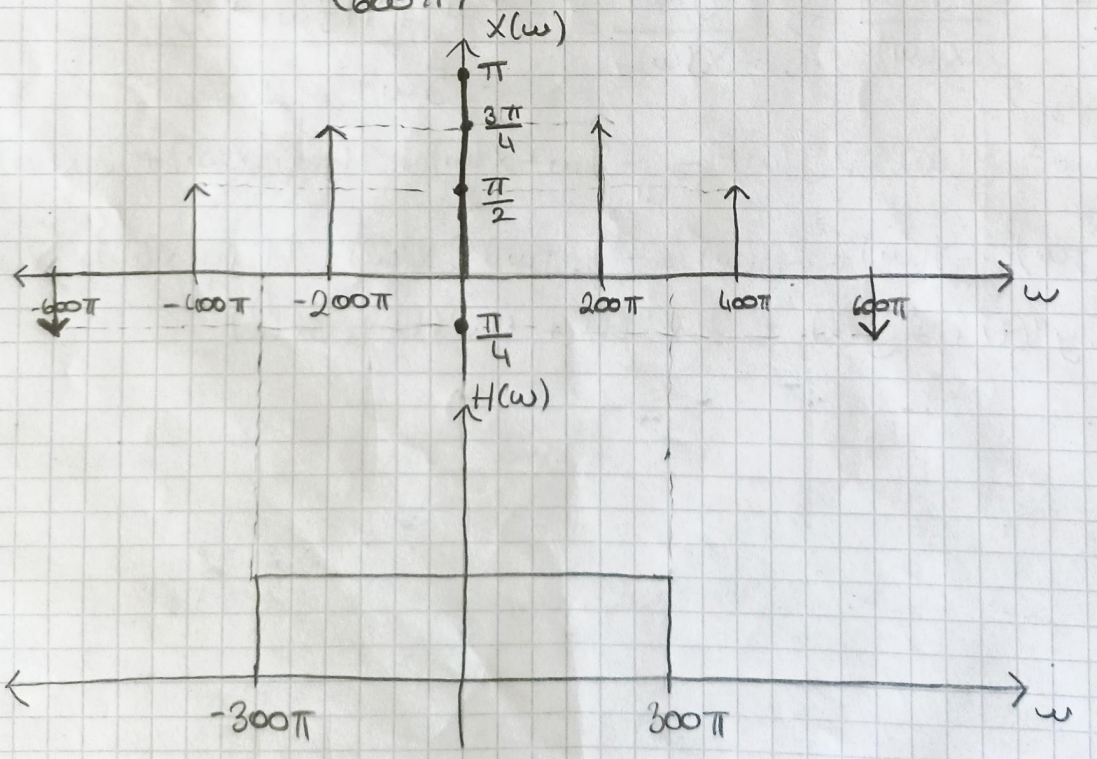
$y(t) = ?$

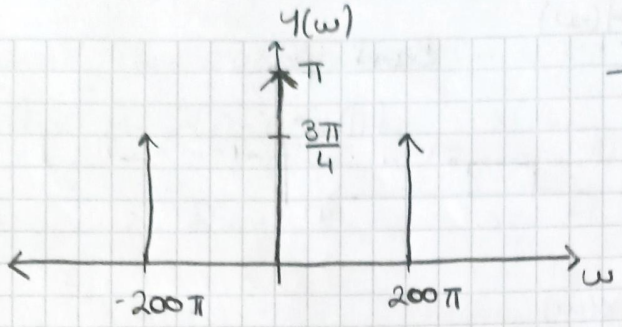
$$X(\omega) = \frac{1}{2} \cdot 2\pi \cdot \delta(\omega) + \frac{3}{4}\pi \cdot \left(\delta(\omega-200\pi) + \delta(\omega+200\pi)\right) + \frac{\pi}{2}\left(\delta(\omega-400\pi) + \delta(\omega+400\pi)\right) - \frac{\pi}{4}\left(\delta(\omega-600\pi) + \delta(\omega+600\pi)\right)$$

$$\text{sinc}\left(\frac{t}{2}\right) \xleftrightarrow{FT} 2\pi\,\text{rect}(\omega)$$

$$\text{sinc}\left(\frac{600\pi t}{2}\right) \xleftrightarrow{FT} \frac{2\pi}{600\pi} \cdot 300 \cdot \text{rect}\left(\frac{\omega}{600\pi}\right)$$

$$H(\omega) = \text{rect}\left(\frac{\omega}{600\pi}\right)$$





$$\pi + \frac{3\pi}{4}\left[\delta(\omega-200\pi)+\delta(\omega+200\pi)\right]$$

$$\boxed{y(t) = \frac{1}{2} + \frac{3}{4}\cos(200\pi t)}$$

Örnek: $h(t) = \frac{2}{\pi} \cdot \text{sinc}(t) \cdot \cos(4t)$

$$x(t) = -1 + 2\cdot\cos(2t) + \cos(4t) - \cos(6t)$$

$y(t) = ?$

$$X(\omega) = -1 \cdot 2\pi\,\delta(\omega) + 2\cdot\pi\cdot\left[\delta(\omega+2)+\delta(\omega-2)\right] + \pi\cdot\left[\delta(\omega+4)+\delta(\omega-4)\right]$$

$$-\pi\left[\delta(\omega+6)+\delta(\omega-6)\right]$$

$$\text{sinc}\,t \xleftrightarrow{FT} \pi\cdot\text{rect}\frac{\omega}{2} = Q(\omega)$$

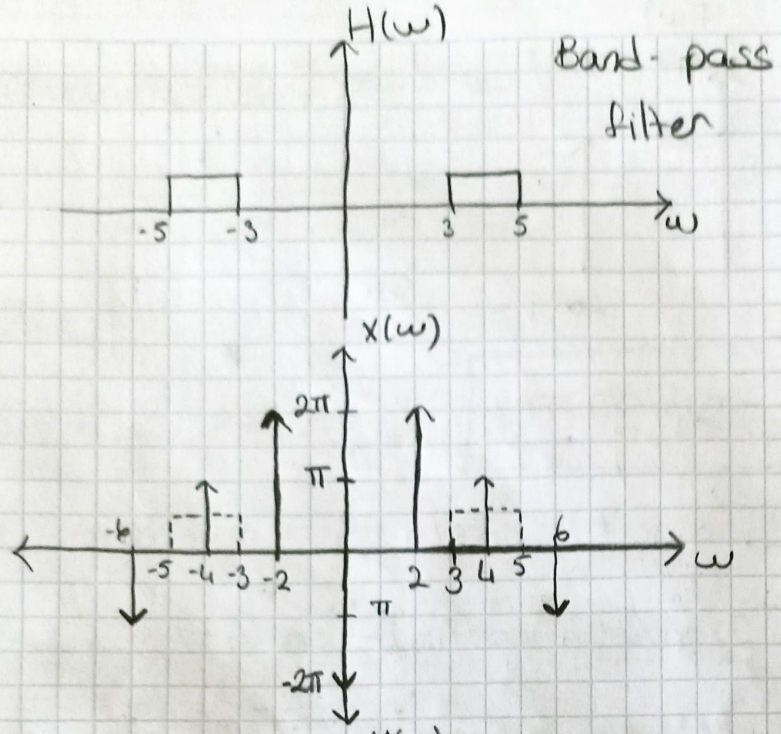
$$h(t) = \frac{2}{\pi}\,\text{sinc}\,t\cdot\left[\frac{e^{j4t}+e^{-j4t}}{2}\right]$$

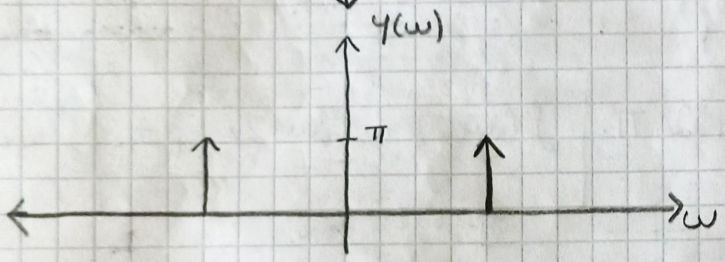
$$h(t) = \frac{1}{\pi}\cdot\text{sinc}\,t\cdot e^{j4t} + \frac{1}{\pi}\cdot\text{sinc}\,t\,e^{-j4t}$$

$$\text{sinc}\,t\cdot e^{j4t} \xleftrightarrow{FT} Q(\omega-4) = \pi\,\text{rect}\left(\frac{\omega-4}{2}\right)\cdot\frac{1}{\pi}$$

$$\text{sinc}\,e^{-j4t} \xleftrightarrow{FT} Q(\omega+4) = \pi\cdot\text{rect}\left(\frac{\omega+4}{2}\right)\cdot\frac{1}{\pi}$$

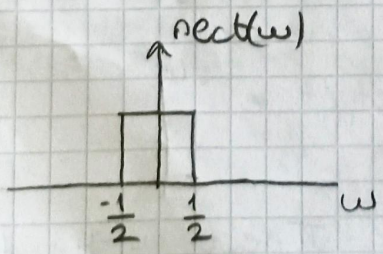
$$H(\omega) = \text{rect}\left(\frac{\omega-4}{2}\right) + \text{rect}\left(\frac{\omega+4}{2}\right)$$

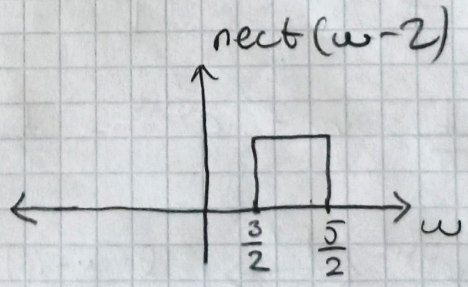


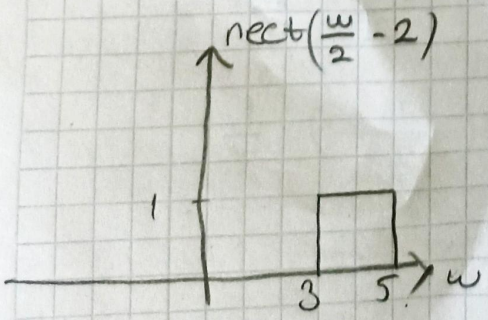


$$Y(\omega) = \frac{1}{\pi}\left[f(\omega - 4) + f(\omega + 4) \right]$$

$$y(t) = 1 \cdot \cos(4t)$$







$a^n u(n) \xleftrightarrow[\text{IDTFT}]{\text{DTFT}} ?$

Ayrık fourier dönüşümü her zaman 2π ile periyodiktir

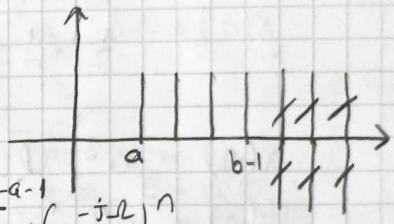
Ayrık Zamanlı Fourier Dönüşümü

Örnek: $x(n) = A \cdot \delta(n-n_0)$, $X(\Omega)$, $X(e^{j\omega})$, $X(e^{j\Omega})$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} A \cdot \delta(n-n_0) \cdot e^{-j\Omega n} = A \cdot e^{-j\Omega n}\Big|_{n=n_0} = A \cdot e^{-j\Omega n_0}$$

Örnek: $x(n) = u(n-a) - u(n-b)$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\Omega n} \Rightarrow \sum_{n=a}^{b-1} \left(e^{-j\Omega}\right)^n$$



$n = n+a$ için $\sum_{n=0}^{b-a-1} \left(e^{-j\Omega}\right)^{n+a} = e^{-j\Omega a} \sum_{n=0}^{b-a-1} \left(e^{-j\Omega}\right)^n$

$$= e^{-j\Omega a}\left(\frac{1-\left(e^{-j\Omega}\right)^{b-a}}{1-e^{-j\Omega}}\right) = e^{-j\Omega\left(\frac{a+b-1}{2}\right)}\left(\frac{\sin\left(\frac{(b-a)\Omega}{2}\right)}{\sin\left(\frac{\Omega}{2}\right)}\right)$$

Örnek: $x(n) = a^n u(n)$

$$X(\Omega) = \sum_{n=0}^{\infty} a^n \cdot e^{-j\Omega n} = \sum_{n=0}^{\infty} \left(a \cdot e^{-j\Omega}\right)^n = \frac{1}{1-a \cdot e^{-j\Omega}}$$

Örnek: $x(n) = a^{|n|}$, $|a| < 1$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} a^{|n|} \cdot e^{-j\Omega n} \Rightarrow \underbrace{\sum_{n=-\infty}^{-1} a^{-n} \cdot e^{-j\Omega n}}_{n=-n \text{ için}} + \sum_{n=0}^{\infty} a^n \cdot e^{-j\Omega n}$$

$$= \sum_{n=1}^{\infty} a^n \cdot e^{j\Omega n} + \sum_{n=0}^{\infty} a^n \cdot e^{-j\Omega n}$$

$$= \sum_{n=1}^{\infty} \left(a \cdot e^{j\Omega}\right)^n + \sum_{n=0}^{\infty} \left(a \cdot e^{-j\Omega}\right)^n = \frac{1}{1-a \cdot e^{j\Omega}} - 1 + \frac{1}{1-a \cdot e^{-j\Omega}}$$

$$= \frac{a \cdot e^{j\Omega}}{1-a \cdot e^{j\Omega}} + \frac{e^{j\Omega}}{e^{j\Omega}-a} = X(\Omega) = \frac{1-a^2}{1-2a \cdot \cos\Omega + a^2}$$

Özel dönüşüm çiftleri

Örnek: $x(n) = 2 \cdot \delta(n) - u(n)$

$$X(\Omega) = 2 - \frac{1}{1-e^{-j\Omega}} - \pi \cdot \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

Örnek: $x(n) = a^n \cdot u(n-3)$ $\quad X(\Omega) = ?$

$$x(n) = a^3 \cdot \underbrace{a^{n-3} \cdot u(n-3)}_{u(n-3)}$$

$$x(n) = a^3 \cdot u(n-3)$$

$$u(n) = a^n \cdot u(n) \longleftrightarrow U(\Omega) = \frac{1}{1-a \cdot e^{-j\Omega}}$$

$$a^3 \cdot u(n-3) \longleftrightarrow a^3 \cdot e^{-j3\Omega} \cdot U(\Omega)$$

$$X(\Omega) = \frac{a^3 \cdot e^{-j3\Omega}}{1 - a \cdot e^{-j\Omega}}$$

Örnek: $x(n) = \cos(\Omega_0 n)$

Çarpma
downsampling
konvolüsyon özelliği

Örnek: $a^{|n|}$, $|a| < 1$

Zamanda ters çevirme

$$x(n) = \underbrace{a^{-n} \cdot u(-n)}_{u_1(-n)} + \underbrace{a^n u(n)}_{u_2(n)} - \delta(n)$$

$$u_2(n) = a^n u(n) \longleftrightarrow U_2(\Omega) = \frac{1}{1 - a \cdot e^{-j\Omega}}$$

$$u_1(-n) = a^{-n} u(-n) \longleftrightarrow U_1(-\Omega) = \frac{1}{1 - a \cdot e^{j\Omega}}$$

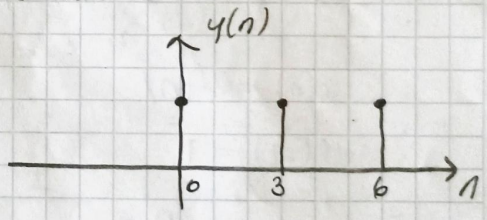
$$X(-\Omega) = U_1(-\Omega) + U_2(\Omega) - 1 = \frac{1 - a^2}{1 - 2a \cdot \cos\Omega + a^2}$$

Örnek: $x(n) = u(n)$

Upsampling
zamanda ölçekleme

$$u(n) \xleftrightarrow{DTFT} \frac{1}{1 - e^{-j\Omega}} + \pi \cdot \sum_{k=-\infty}^{\infty} \delta(-\Omega - 2\pi k)$$

$$y(n) = (\uparrow 3)\, x(n) = x\left(\frac{1}{3} n\right)$$



$$Y(\Omega) = X(3\Omega)$$

$$Y(\Omega) = \frac{1}{1 - e^{-j3\Omega}} + \pi \cdot \sum_{k=-\infty}^{\infty} \delta(3\Omega - 2\pi k)$$

Örnek: $x(n) = n \cdot a^n u(n)$

$w(n) = a^n u(n)$, $x(n) = n \cdot w(n)$

$$W(\Omega) = \frac{1}{1 - a \cdot e^{-j\Omega}}$$

$$X(\Omega) = j \cdot \frac{d\, W(\Omega)}{d\Omega} = j \cdot \frac{-\left(-a \cdot (-j \cdot e^{-j\Omega})\right)}{\left(1 - a\, e^{-j\Omega}\right)^2}$$

$$= \frac{a \cdot e^{-j\Omega}}{\left(1 - a \cdot e^{-j\Omega}\right)^2}$$

Örnek: $x(n) = u(n)$, $u(n) = \sum_{k=-\infty}^{n} f(k)$

$w(n) = f(n)$

$W(\Omega) = 1$

$$x(n) = \sum_{k=-\infty}^{n} w(n) \longleftrightarrow X(\Omega) = \frac{W(\Omega)}{1 - e^{-j\Omega}} + \pi \cdot W(0) \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

Örnek: $x(n) = n \cdot \left(\frac{1}{2}\right)^{|n+3|}$, $X(\Omega) = ?$

$$w_1(n) = \left(\frac{1}{2}\right)^{|n|} \longleftrightarrow W_1(\Omega) = \frac{1 - \frac{1}{4}}{1 - 2\cdot\frac{1}{2}\cdot\cos\Omega + \frac{1}{4}} = \frac{3}{5 - 4\cos\Omega}$$

$$w_2(n) = w_1(n+3) \longleftrightarrow W_2(\Omega) = e^{j3\Omega} \cdot W_1(\Omega) = \frac{3 \cdot e^{j3\Omega}}{5 - 4\cdot\cos\Omega}$$

$$x(n) = n \cdot w_2(n) \longleftrightarrow X(\Omega) = j\,\frac{d\,W_2(\Omega)}{d\Omega}$$

$$X(\Omega) = j\left(\frac{j9\cdot e^{j3\Omega}\cdot(5 - 4\cos\Omega) - 12\cdot\sin\Omega\, e^{j3\Omega}}{(5 - 4\cos\Omega)^2}\right)$$

Örnek: $x(n) = (n+1)\cdot a^n\, u(n)$, $X(\Omega) = ?$

$$x(n) = n\cdot a^n\cdot u(n) + a^n\cdot u(n)$$

$$X(\Omega) = j\,\frac{d}{d\Omega}\,\frac{1}{1 - a\cdot e^{-j\Omega}} + \frac{1}{1 - a\cdot e^{-j\Omega}} = \frac{1}{(1 - a\cdot e^{-j\Omega})^2}$$

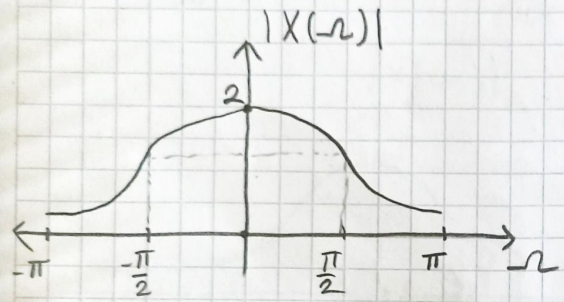
Örnek: $y(n) = \sum_{k=-\infty}^{\infty} e^{-j\Omega k} \cdot x(-k)$,

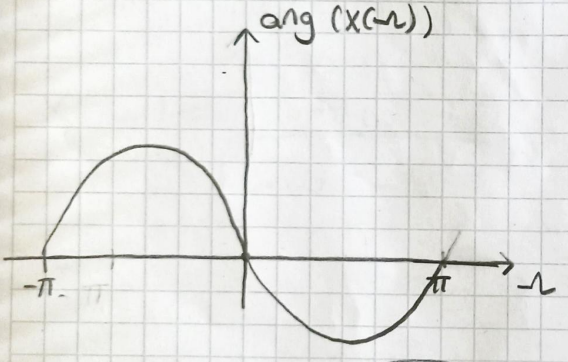
$x(n) \xleftrightarrow{FT} X(\Omega)$

$y(\Omega) = ?$

Örnek: $x(n) = \left(\frac{1}{2}\right)^n u(n)$ Frekans spektrumunu belirleyiniz.

$$X(\Omega) = \frac{1}{1 - \frac{1}{2} \cdot e^{-j\Omega}} = \frac{1}{\frac{2 - e^{-j\Omega}}{2}} = \frac{2}{2 - e^{-j\Omega}}$$





$$|X(\Omega)| = \frac{|2|}{|2 - e^{-j\Omega}|}$$

$$|X(\Omega)| = \frac{|2|}{|2 - \cos\Omega + j\sin\Omega|}$$

$$= \frac{2}{\sqrt{(2 - \cos\Omega)^2 + \sin^2\Omega}}$$

$$= \frac{2}{\sqrt{5 - 4 \cdot \cos\Omega}}$$

$$\arg X(\Omega) = \underbrace{\arg(2)}_{0} - \arg\left(2 - \cos\Omega + j\sin\Omega\right)$$

$$- \arctan\left(\frac{\sin\Omega}{2 - \cos\Omega}\right)$$

Örnek: $h(n) = u(n+5) - u(n-6)$ Frekans tepkisi nedir?

$H(\Omega) = ?$

$$H(\Omega) = e^{j5\Omega}\left[\frac{1}{1-e^{-j\Omega}} + \pi\cdot\delta(\Omega)\right] - e^{-j6\Omega}\left[\frac{1}{1-e^{-j\Omega}} + \pi\delta(\Omega)\right]$$

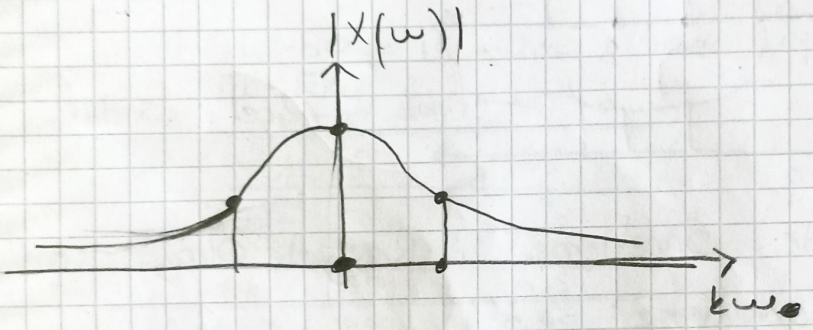
$$h(n-5) = u(n) - u(n-11)$$

$$\left(1 - e^{-j11\Omega}\right) u(\Omega)$$

$$X(\omega) = \frac{4}{(j\omega + 2)(j\omega - 2)}$$

$$\frac{4}{j\omega^2 - 2^2} = \frac{-4}{\omega^2 + 4}$$

$$\sqrt{\left(\frac{-4}{\omega^2+4}\right)^2} = \sqrt{\frac{16}{(\omega^2+4)^2}} = \frac{4}{|\omega^2+4|}$$



$$X(\omega) = \frac{+4}{-\omega^2 - 4} = \frac{4}{j\omega^2 - 4} = \frac{4}{-|4 + j\omega^2|}$$

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2 dakika film

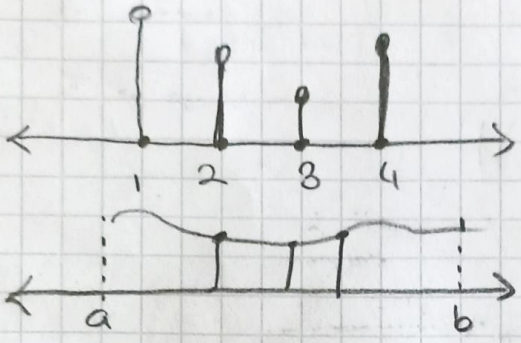
Olasılık ve istatistik

Popülasyon: evren, nüfusu

Parametre: Popülasyonla ilgili sayısal bilgi (ortalama, standart sap.)

İstatistik: Örneklem ile ilgili sayısal bilgi (ortalama, standart sap.)

Nicel (sayısal) veriler:



Nicel ayrık

Nicel sürekli

Örnekleme: Rastgele örnekleme, Rastgele olmayan örnekleme

Rastgele olmayan örnekleme:

- Kolay örnekleme: Lise cd.sinde bekleyip birilerine soru sormak
- Kota örnekleme: Belirli özelliklere göre örneklem oluşturma

---

Bias: Seçilen örneklemin yanlı olması durumu (eğilim)

Küme ve tabaka arasındaki fark

↳ Homojen (Her tabakadan eşit oranda seçim)

Plasebo etkisi: Bir kişinin aldığı işlemin etkili olduğuna inanması nedeniyle yanıt değişkeninde değişim görülmesidir.

Körleme yöntemi:

Pareto grafiği !

Kök - yaprak grafiği

55, 58, 73, 84, 95

Orta miktarda veriler için kullanılır.

Hangi verinin az veya çok değiştiği önemlidir.

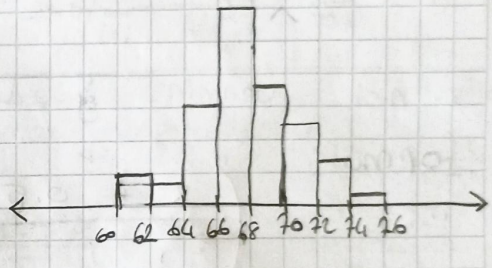
kök	yaprak	frekans
5	5, 8	2
6		0
7	3	1
8	4	1
9	5	1

Aralıklı tablolar oluşturma → dağılım aralığı

74 - 60 = 14 ÷ 2 = 7 sınıf sayısı

FB - EK → dahil, değil

	f	nf
60 - 62	5	%05
62 - 64	3	% 3
64 - 66	15	%15
66 - 68	40	% 40
68 - 70	17	% 17
70 - 72	12	% 12
72 - 74	7	% 7
74 - 76	1	% 1



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1 3 3 [4] 4 6 7 → medyan

$M = \left(\frac{n+1}{2}\right)$. veri

2 2 [3 | 4] 4 6 → medyan = (4+3)/2 = 3.5

Medyan için verinin sıralı olması gerekir.

114.950, 158.000, 230.500, 387.000, 389.950, 479.000

488.800, 529.000, 575.000, 639.000, 659.000, 1095.000, 5500.00

$M = 488.800$

$Q_1 = \frac{230.500 + 387.000}{2} = 308.750$

$Q_3 = \frac{639.000 + 659.000}{2} = 649.000$

$IQR = Q_3 - Q_1 = 649.000 - 308.750 = 510.375$

Boxplot önemli Q_1, Q_3, IQR ööö

70. yüzdeliği bulmak için; $i = \frac{k}{100}(n+1)$

$\frac{k}{100} \cdot (n+1)$ → $29+1$ → veri sayısı

$\frac{70}{100} \cdot 30 = 21.$ veri

83. yüzdeliği bulun;

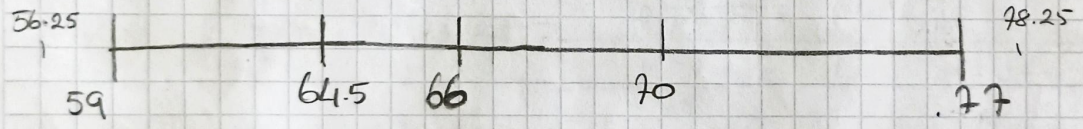
$\frac{83}{100} \cdot 30 = 24.9$ 24 ve 25. verinin ortalaması

$19 = \frac{k}{100} \cdot 30$ $\quad k = \frac{1900}{3}$

x.'nci verinin yüzdelik değerini bulmak için formül:

y: tekrar sayısı (frekans)

$$\left(\frac{x + 0.5y}{n}\right) \times 100$$



$IQR = 70 - 64.5 = 5.5$

$1.5 \times IQR = 8.25$

$Q_1 - IQR = 64.5 - 8.25 = 56.25$

$Q_3 + IQR = 70 + 8.25 = 78.25$

Merkezi Eğilim Ölçüleri:

Ortalama ve medyan karşılaştırılması genelde anlamlıdır.

$\bar{x}$: Örneklem ortalaması

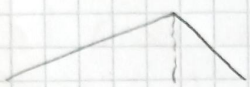
μ: popülasyon ortalaması

Mod: en çok tekrar eden veri (birden fazla mod olabilir)

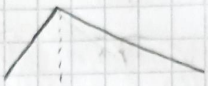
Chebyşev kuralı

Çarpıklık

Histogram grafiği gibi düşünelim.



sola çarpık



sağa çarpık

Standart sapma

Örneklem için

$$s = \sqrt{\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}}$$

Popülasyon için:

$$\sigma = \sqrt{\sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}}$$

Z - değeri farklı örneklemleri kıyaslamak için kullanılır.

$$Z = \frac{x - \bar{x}}{s} \quad , \quad Z = \frac{x - \mu}{\sigma}$$

	Not ort	Okulun not ort	Okul sap. Standart
Ahmet	2.85	3.0	0.7
Ali	77	80	10

$$Z_{Ahmet} = \frac{x - \mu}{\sigma} \Rightarrow \frac{2.85 - 3.0}{0.7} = -0.21$$

$$Z_{Ali} = \frac{x - \mu}{\sigma} \Rightarrow \frac{77 - 80}{10} = -0.3$$

Z değeri büyük olan daha iyi durumdadır

Olabilirlik (Likelihood)

Olasılık (Probability)

Gitti, bin deneyin sonucudur.

Örnek uzayı, tüm ihtimallerin kümesi

$S = \{Y, T\}$ para deneyi örneği

$A = \{Y\} \longrightarrow$ A olayı

$B = \{T\} \longrightarrow$ B olayı

$P(A) \rightarrow$ A olayının gerçekleşme olasılığı

ve

$$P(A) = \frac{\text{A olayının meydana gelme sayısı}}{\text{Toplam deneme sayısı}} \quad 0 \leq P(A) \leq 1$$

(Deneysel olasılık) — istatistiksel model

Toplam deneme sayısını arttırdıkça teorik sonuçlara daha yakınlaşırız.

A veya B

$A \vee B$

$A \cup B$

A ve B

$A \wedge B$

$A \cap B$

$AB = BA$

$A' \rightarrow A$'nın tümleyeni

$A = \{1,2,3\}$, $S = \{1,2,3,4,5\}$

$A' = \{4,5\}$

$P(A) + P(A') = 1$

Koşullu olasılık

$P(A \mid B)$ veya $P(A \mid B) \longrightarrow$ B'den sonra A olayının gerçekleşme olasılığı

$$P(A \mid B) = \frac{P(AB)}{P(B)} = \frac{P(BA)}{P(B)}$$

BAYES

$S = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4)\}$ Örnek uzayı

$A = \{(1,1), (1,3), (2,2), (2,4), (3,1), (3,3)\}$ Toplamların çift olması

$B = \{(2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4)\}$ İlk sayının asal olması

$P(G \setminus H) \stackrel{?}{=} P(G)$

$P(H \setminus G) \stackrel{?}{=} P(H)$

$P(G) = 0.6$

$P(H) = 0.5$

$P(GH) = 0.3$

$P(G \setminus H)$

Hilesiz iki para havaya atılıyor. Cevaplandıralım

a) F = en fazla bir yazı gelme olayı

$S = \{YY, YT, TY, TT\} : 4$

$F = \{YT, TY, TT\} : 3$

b) iki paranın aynı olma olayı = G

$G = \{YY, TT\} : 2$

c) H = ilk paranın tura, ikinci paranın yazı veya tura gelmesi

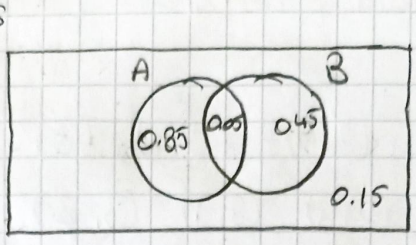
$H = \{TY, TT\} : 2$

d) J = her iki paranın yazı gelme olayı.

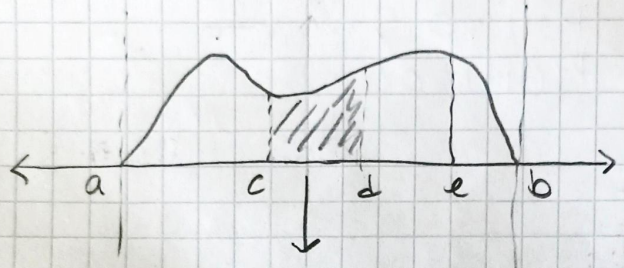
$J = \{YY\} : 1$

$P(A) = 0.4$ $P(AB) = 0.05$

$P(B) = 0.5$



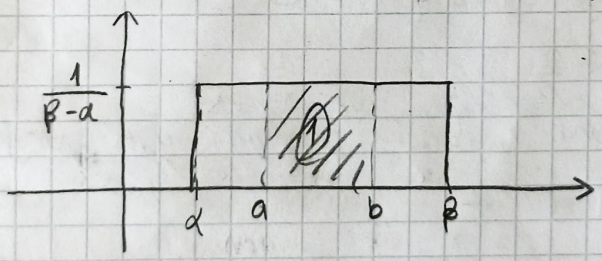
Sürekli Dağılım Fonksiyonları



$P(c < X < d)$ ve $P(a < X < b) = 1$

$P(X = e) = 0$ $\int_e^e f(x)\,dx = 0$

Düzgün Dağılım



$(\beta - \alpha) \cdot \frac{1}{\beta - \alpha} = 1$

$P(a < X < b) = \frac{b-a}{\beta - \alpha}$

$$E(X) = \int_\alpha^\beta x \cdot \frac{1}{\beta-\alpha}\,dx = \frac{1}{\beta-\alpha} \left.\frac{x^2}{2}\right|_\alpha^\beta = \frac{1}{\beta-\alpha}\left(\frac{\beta^2 - \alpha^2}{2}\right)$$

$$= \frac{1}{\beta-\alpha} \cdot \left(\frac{(\beta-\alpha)\cdot(\beta+\alpha)}{2}\right) = \frac{\beta+\alpha}{2}$$

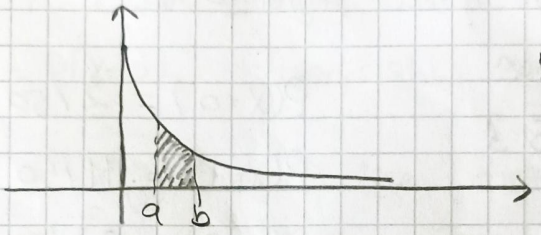
Dağılımın ne üzerine olduğu ve parametreler

Üstel Dağılım

$$\lambda = \frac{1}{\alpha}$$

$$f(x) = \begin{cases} \alpha \cdot e^{-\alpha x} & , x > 0 \\ 0 & , \text{diger} \end{cases}$$

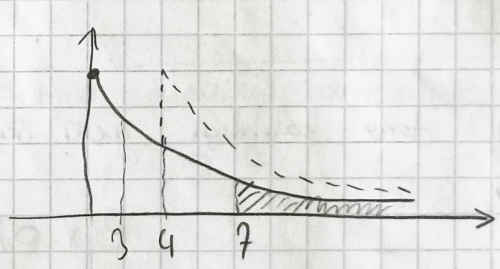
$$E[X] = \frac{1}{\alpha} \qquad Var(X) = \frac{1}{\alpha^2}$$



$$P(a < X < b) = P(X < b) - P(X < a)$$

$\lambda = 4$

$\alpha = \frac{1}{4} = 0.25$



Örnek: Bir yılda ortalama kaza sayısı : 150
Analiz ayında 10 kaza meydana gelme olasılığı

$P(X = 10)$

$\lambda \neq 150$

$\lambda = \frac{150}{12} = 12.5$, 1 ayda ortalama kaza

↳ incelenen veri aylık ortalama kaza olduğu için yıllık ortalama kaza sayısı bizim için λ değeri olmaz.

Ayrık Rastgele Değişkenler

Olayların sayısal değerlerle eşleşmesine rastgele değişken denir.

Örneğin: $S = \{ Tura, Yazı \}$

$X(Tura) = 1 \qquad X(Yazı) = 0$ olsun

$P(X=1) = 0.5 \qquad P(X=0) = 0.5$

X: Bebeğin annesini uyandırma sayısı olsun

x: X'in alacağı değerler

$x = \{ 0, 1, 2, 3, 4, 5 \}$

$P(X=0) = 2/150 \qquad P(X=3) = 9/150$

$P(X=1) = 11/150 \qquad P(X=4) = 4/150$

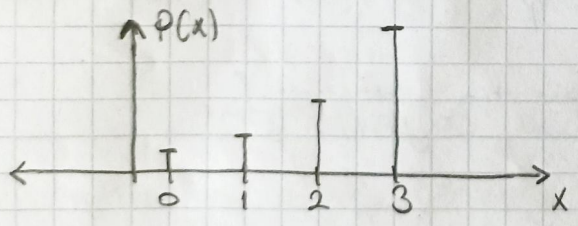
$P(X=2) = 23/150 \qquad P(X=5) = 1/150$

X: Nancy'nin derse katıldığı gün sayısı

$x = \{ 0, 1, 2, 3 \}$

$P(X=0) = 1/100 \qquad P(X=2) = 15/100$

$P(X=1) = 4/100 \qquad P(X=3) = 80/100$



Olasılık kütle fonksiyonu

$$\binom{x}{n} = \frac{x!}{n! \cdot (x-n)!}$$

Beklenen değer

$$E(X) = \mu = \sum_i x_i \cdot P(x = x_i)$$

Hilesiz zar için 3.5 değeri beklenen değerdir.

Binom Dağılımı

$n=1$ ise Bernoulli deneyi de denir.

$x = \{0, 1\}$ kullanımının avantajları vardır.

$p = P(x=1)$: başarı olasılığı

$q = 1-p = P(x=0)$: başarısızlık olasılığı

Binom örnek1 : X: yolcunun lise mezunu olması

$n=8$ $\quad q = 0.2 \quad p = 0.8$

Binom örnek2: X: öğrencinin ödevini zamanında yapmış olması

$X \sim B(n, p)$ $\quad n = 50 \quad p = 0.7 \quad q = 0.3$

$P(X \geq 40)$

n değeri büyüdükçe sürekli dağılıma yaklaşır.

Geometrik Dağılım

$$X \sim G(p) \qquad E(X) = \frac{1}{p} \qquad \sigma^2 = \frac{1-p}{p^2}$$

Örnek1: Arızalı olma olasılığı başarı olarak kabul edilir.
(Hatalı sınırı bulmak)

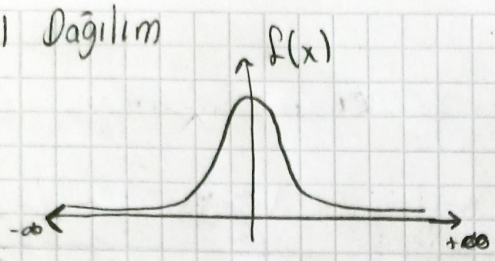
$p = 0.02 \qquad P(X=7)$

$$P(X=7) = (1-p)^{7-1} \cdot p = (0.98)^6 \cdot (0.02) = 0.0177$$

$$E(X) = \frac{1}{p} = \frac{1}{0.02} = 50$$

Arızalı bir parçaya denk gelmek için ortalama 50 ürün test edilmelidir.

Normal Dağılım



Normal dağılım PDF

$X \to N(\mu, \sigma)$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$z = \frac{x - \mu}{\sigma}$$ dönüşümü

Örnek 1: 25 ort 4 standart sapma

$P(25 < X < 32) = ?$ $\quad z = \frac{x-\mu}{\sigma}$ dönş.

$P(18 < X < 34) = ?$

$$P\left(\frac{25-25}{4} < z < \frac{32-25}{4}\right) = P(z<0) -$$

Merkezi Limit Teoremi

Bir popülasyondan k adet örneklem seçilsin

$\bar{X}_1, \bar{X}_2, \ldots, \bar{X}_k$ normal dağılım sergiler

Örneklem boyutunun $(n) > 30$ olması yeterli kabul edilir.

Örnek 1: $E(x) = 8$

$Var(x) = 4$

$n = 49 \quad \sigma = 2$ dk

$\mu_x = 8$ dk

$P(7.8 < \bar{X} < 8.4)$

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}} = \frac{2}{7}$$

$$\mu_{\bar{x}} = \mu_x = 8$$

$$P\left(\frac{7.8-8}{2/7} < z < \frac{8.4-8}{2/7}\right)$$

$$P(-0.7 < z < 1.4)$$

Örnek 2: $\mu = 34 \quad \sigma = 15 \quad n = 100$

$$\mu_{\bar{x}} = \mu = 34 \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = 1.5$$

$P(\bar{X} > 30) = ?$

$$P\left(z > \frac{30-34}{1.5}\right) = P(z > 2.67) = 0.9962$$

$$= 1 - P(z < 2.67) = 0.0038$$

Toplamlar için CLT

$\mu = 90 \qquad \sigma = 15 \qquad n = 80$

$$P(\Sigma x > 7500) = P\left(Z > \frac{7500 - \mu_{\Sigma x}}{\sigma_{\Sigma x}}\right) = P(Z > 2.236)$$

$\mu_{\Sigma x} = n \cdot \mu = 7200$

$\sigma_{\Sigma x} = \sqrt{n} \cdot \sigma = 134.1$

n değeri sonsuza yaklaştığı zaman binom dağılımı, normal dağılıma yakınsar ve normal dağılıma yakınsamak için

Temel ⇒ $n \cdot p \geq 5$ ve $nq \geq 5$

$p = 0.46$

$q = 0.54$

$X \sim B(n, p) \Rightarrow X \sim N(230, 11.145^2)$ 124,2

$\mu = n \cdot p = 230$

$\sigma = \sqrt{npq} = 11.144$

$$P\left(Z > \frac{249.5 - 230}{11.145}\right) = P(Z > 1.75) = 0.0401$$

Güven Aralıkları karar teorisi $\boxed{a < \mu < b}$, %99 doğru

$X \sim N(\mu_{\bar{x}}, \sigma_{\bar{x}}) \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

$\bar{x} - \mu = \varepsilon$

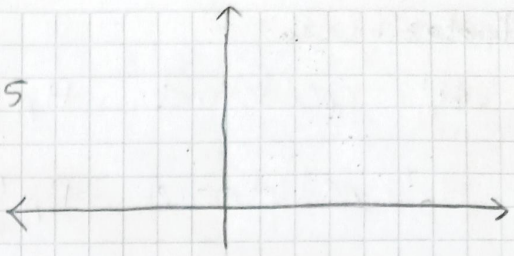
Güven aralığı hesaplarken $n \geq 30$ ve populasyon varyansı bilinmelidir.

Örnek 1:

$\sigma = 2.4$ cm

$n = 16$ $\quad \alpha = 0.05$

$\bar{X} = 3.2$ cm



X: Parçaların dış çapı (cm)

$EBM = z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} = z_{\frac{\alpha}{2}} \cdot \frac{2.4}{4}$

$P(z_{\frac{\alpha}{2}}) = 1 - \frac{\alpha}{2} = 0.975$

$z_{\frac{\alpha}{2}} = 1.96 \Rightarrow EBM = 1.176$

%95 güven aralığı $\quad 2.024 < \mu < 4.376$

veya $(2.024, 4.376)$

---

t dağılımı

$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$

2.797

! Doğadaki birçok şey normal dağılır.

! popülasyon dağılım türü önemli değildir. $n \geq 30$ için kullanılabilir ve σ bilinmiyor ise kullanılır.

Hipotez testi

H_0'ı reddetmek H_A'yı kabul etmek anlamına gelir.

H_0'ı reddetmemek H_A'yı kabul etmemek anlamına gelir.

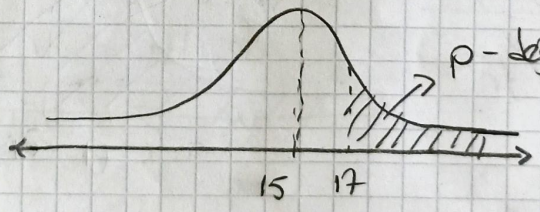
kahve araştırması:

H_0: kahve, kanser riskini artırmıyor (ilk akla gelen)

H_A: kahve, kanser riskini artırmıyor. (test edilecek durum)

	H_0 doğru	H_0 yanlış
H_0'ı reddetmeme	Doğru karar	Tip II hata β
H_0'ı reddet	Tip I hata α	Doğru karar

Hipotez testi ve örnekleme - Örnek



$$P(\bar{X} > 17) = P\left(Z > \frac{17-15}{0.5/\sqrt{10}}\right) = P(Z > 12.65) = 0$$

H_0'ı reddet

$p < \alpha$ ise H_0'ı reddet

$p > \alpha$ ise H_0'ı reddetmeyin

Karar ve Sonuç - Örnek:

olasılık ↑

p değeri > α

dolayısıyla H_0 kabul et

H_0

$H_0 : p = 0.5$ $\quad \alpha = 0.01$

$H_A : p > 0.5$ $\quad$ p değeri = 0.025

Problem - 1

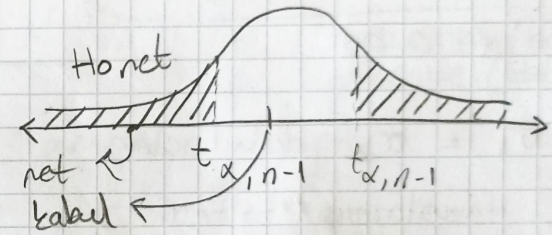
$\bar{X}$: 71.8 yıl

Örneklem boyutu α ve β üzerinde ne kadar etkilidir.

Problem - 2

$H_0 : \mu \leq 46$ $\quad \alpha = 0.05$

$H_A : \mu < 46$



Hipotez Testi (2 örneklem ile)

Örnek: İki farklı tedavi yönteminin ortalama iyileşme süreleri

$\sigma_1 = \sigma_2$ ise havuzlanmış t-testi (çok nadir)

$\sigma_1 \neq \sigma_2$ ise Welch t-testi (genel durum)

Welch t-testi

Test istatistiği formülü:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Serbestlik derecesi hesaplarken değişken değiştirme yap.

İki örneklemli hipotez testi adımları

1. Hipotezleri belirle
2. Anlamlılık düzeyini seç
3. Test istatistiğini hesapla
4. Karar kuralını belirle
5. Karar verme
6. Sonucu yorumlama

Problem - 1

→ popülasyonlar yaklaşık normal dağılır. deniyorsa örneklem boyutu önemli değil

$n_1 = 12$ $n_2 = 10$

$\alpha = 0.05$

$\bar{X}_1 = 85$ $s_1 = 4$

$\bar{X}_2 = 81$ $s_2 = 5$

$\sigma_1 = \sigma_2$ kabul edilir

Havuzlanmış t-testi

$H_0 : \mu_1 - \mu_2 = 2$

$H_A : \mu_1 - \mu_2 > 2$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{12} + \frac{1}{10}}} = 1.04$$

↳ Virgülden sonra 4 basamak

σ değerleri bilindiğinde

serbestlik derecesine gerek yoktur.

Chi-square (ki-kare dağılımı) (χ^2)

* Veriler arasındaki ilişkiyi anlamak

χ^2 tablosu vardır ve serbestlik derecesi bulunur (n)

Matematiksel tanım

$$\chi^2 = z_1^2 + z_2^2 + \dots + z_k^2$$

$df = k-1$

$$z_i \sim N(0,1)$$

Standart sapma: $\sqrt{2 \cdot df}$ μ: d.f

$$\chi^2 = \sum_{i=1}^{k} \frac{(o_i - e_i)^2}{e_i}$$

e_i: expected (beklenen f)
o_i: observe (gözlenen f)

$e_i > 5$ olması beklenir. (değilse gruplama yapılır)

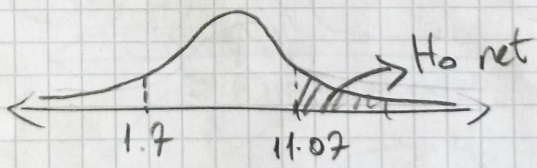
Örnek1: H_0: düzgün dağılıma sahiptir $df = 5$ $\mu = 5$

$\alpha = 0.05$ $SS = \sqrt{10}$

$\chi^2 = 1.7$

karar: $\chi^2 < \chi^2_{\alpha, df}$ olduğu için H_0'ı reddetmeyiz

$$\chi^2_{\alpha, df} = \chi^2_{0.05, 5} = 11.07$$



p-değeri hesabı $>> \alpha$ olduğu için kararımızdan eminiz.

Uyum iyiliği testi

Örnek 2: H_0: X normal dağılıma sahiptir

H_A: sahip değildir.

Bağımsızlık testi

H_0: iki değişken bağımsızdır.

H_A: iki değişken bağımsız değildir.

$df = (r-1).(c-1)$

Örnek 1: $r = 2$ $c = 3$

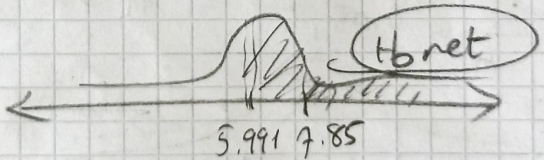
$df = 2$ | $\alpha = 0.05$ kabul edelim | ve $\chi^2_{0.05,2} = 5.991$

Beklenen değer tablosu

200.9	209.9	187.2
135.1	141.1	125.8

$\chi^2 = 7.85$

6 serbestlik derecesi kadar e değeri hesaplamak yeterlidir. (minimum)



Homojenlik Testi

İki veya daha fazla farklı popülasyonun benzerliği

H_0 popülasyonların tümü homojendir.

H_A: popülasyonlar homojen değildir.

Örnek 1: $n = 3$ $c = 3$ $df = 4$ $\alpha = 0.05$

H_0 reddedilemez

Varyans Testi

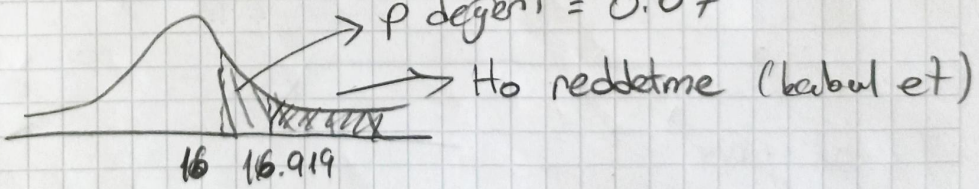
$$\chi^2 = \frac{(n-1) \cdot s^2}{\sigma_0^2} \qquad df = n - 1$$

Popülasyon normal değiliyor ise yapılabilir.

Örnek1: $s: 1.2$ $n = 10$ $\alpha = 0.05$

$\sigma > 0.9$?? $\sigma_0^2 = 0.81$ $df = 9$

$$\chi^2 = \frac{9 \times 1.44}{0.81} = 16 \qquad \chi^2_{0.05,9} = 16.919$$



Örnek 2: $n = 20$ $\alpha = 0.05$ $df = 19$

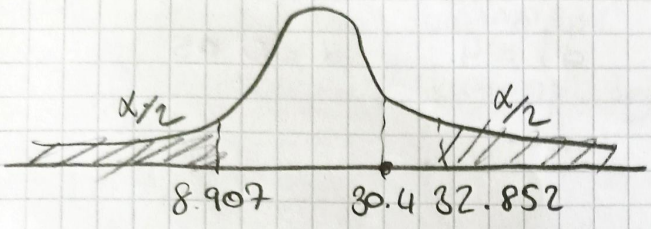
$s^2 = 0.08$ $\sigma_0^2 = 0.05$

$$\chi^2 = \frac{19 \times 0.08}{0.05}$$

$\chi^2_{0.05,19} =$

$\chi^2_{0.025,19} = 32.852$

$\chi^2_{1-\frac{0.025}{2},19} = 8.907$



H_0 red değil (kabul) ✓

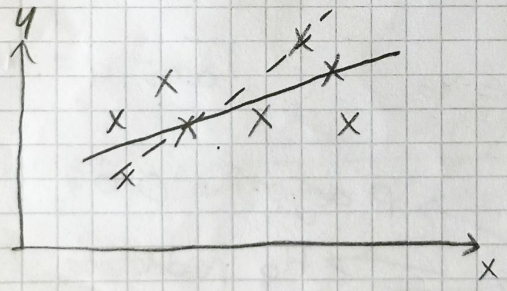
P-değeri kolay hesaplanmaz (simetrik değildir.)

Doğrusal Regresyon ve Korelasyon

Korelasyon, nedensellik anlamına gelmez.

r: örnekten korelasyonu

ρ: popülasyon korelasyonu



Hangi doğru en uygun?

$$y = b_0 + b_1 x$$

$$b_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \qquad b_0 = \bar{y} - b_1 \bar{x}$$